Candidate-pool opportunity and selection stability under finite validation in molecular property prediction: a repeated nested audit

[Author names and ORCID iDs not provided — CONFLICT HOLD]

[Institutional affiliations and full postal addresses not provided — CONFLICT HOLD]

*Corresponding author: [name, email and postal address not provided — CONFLICT HOLD]

Running title: Candidate-pool opportunity under finite validation

Abstract

Background: Molecular property-prediction studies compare increasingly large, correlated registries of representations, learners and tuning variants. Expansion can add complementary chemical information, but finite validation data may not rank the enlarged opportunity set reproducibly.

Methods: We audited nine public endpoints with 32 registered lightweight candidates, K = 4, 8, 16 and 32, three inner and three outer scaffold folds, and ten split seeds. A K-invariant K = 32 leave-one-seed-out reference was primary; K-dependent cross-fitting and the same-fold maximum were sensitivity and descriptive opportunity quantities. We additionally evaluated practical-equivalence grids, PR-AUC and training-defined minority-recall-constrained selection, constructed duplicate/weak/complementary controls, empirical correlated heteroskedastic recovery simulations, registry composition, compute and chemical support.

Results: Against the K-invariant full-registry reference, K = 32 minus K = 4 completion-gap contrasts were negative in 8 of 9 endpoints and positive in 1; negative contrasts mean a smaller full-registry completion gap at K = 32, whereas a positive contrast means a smaller gap at K = 4. Descriptive split-seed sensitivity intervals excluded zero for 7 endpoints. At K = 32, 79.6% of selections met the retrospectively locked task-matched middle practical-equivalence tolerance. PR-AUC and minority-constrained selectors changed the ROC-AUC-selected candidate in 52.7% and 66.0% of classification audit units, respectively; the minority-recall rule did not reliably attain the 0.80 target on held-out folds.

Conclusions: Candidate-pool expansion changed available opportunity and the stability with which finite validation realised it. The result depended on endpoint, registry composition, selection metric, candidate dependence, chemical support and split mechanism. K-invariant cross-fitting, practical-equivalence reporting and metric-matched selection reduced interpretive ambiguity but did not provide external validation or establish a universal effect of larger registries.

Scientific Contribution: This study separates a K-invariant cross-fitted reference from K-dependent and same-fold opportunity quantities, and quantifies practical equivalence and metric-dependent selection in molecular benchmarks. Constructed dependence controls and empirical correlated heteroskedastic simulations show why nominal candidate count cannot substitute for effective search diversity. The resulting audit and reporting package is designed for third-party reproduction rather than architecture ranking.

Keywords: molecular property prediction; candidate-pool expansion; nested cross-validation; cross-fitted reference; practical equivalence; PR-AUC; selection stability; chemical support

1 Introduction

Modern molecular property prediction now encompasses a candidate space extending far beyond a few fixed fingerprints and regressors. Contemporary studies may compare circular and substructure fingerprints, physicochemical descriptor panels, graph models, directed message-passing models, chemical-language models, pretrained representations, automated machine-learning systems, ensembles and post-hoc calibration strategies within the same benchmark [1,2,5–10,16–24,34,35]. Each component can encode a different inductive bias: fingerprints emphasize local motifs, descriptors summarize global molecular properties, graph networks learn neighbourhood interactions, and language models transfer regularities from large molecular corpora. Reported performance therefore depends on both the predictive capacity of individual candidates and the candidates admitted to the search, the extent of tuning, the utility comparisons and the winner-selection rule. These choices can materially alter the resulting benchmark conclusions. As registries grow, model search becomes part of the estimand rather than a neutral preliminary step. Nevertheless, benchmark reports often describe the selected predictor more fully than the selection process that produced it.

Expanding a candidate pool has two distinct effects: genuine representational opportunity arises when an additional representation or learner captures chemical information that the existing registry cannot express; repeated-selection opportunity arises because every eligible candidate contributes another noisy utility estimate that can be ranked using the same finite validation information. The first mechanism can improve generalization by adding complementary chemical signal; the second can favour a candidate whose apparent advantage is specific to the available validation scaffolds. Correlation among candidates moderates but does not remove this tension, because near-duplicate pipelines may share most errors while still exchanging ranks under finite sampling. This effect does not require direct test leakage: repeated comparisons among representations, learners, tuning variants, calibration rules and ensemble choices can consume validation information even when the final audit fold remains hidden during fitting [3,4]. A larger registry may therefore create useful opportunity and additional selection pressure at the same time.

Nested cross-validation is the principal design for separating these stages. Inner folds support candidate ranking, hyperparameter selection and any fitted preprocessing, whereas an outer fold evaluates the frozen decision produced by that inner procedure. This separation prevents the tuning score from serving as the final performance estimate and permits alternative selectors to be compared on shared outer folds. Repeated seeded scaffold partitions further expose sensitivity to chemically coherent test groups and allow paired candidate comparisons under identical audit conditions. For molecular data, where random splits can place closely related compounds on both sides of the evaluation boundary, nested scaffold designs therefore provide an important safeguard. They preserve a clear distinction between inner selection and outer evaluation and provide an auditable record of how a registered selector behaves across repeated chemical partitions [25].

Nested evaluation nevertheless leaves several search properties unresolved. It does not quantify candidate dependence or establish that nominal candidate count equals effective diversity. It does not equalize compute exposure when larger registries require more fits, nor does it automatically account for unavailable or failed candidates. An outer score validly evaluates a frozen selector, but the maximum taken across many outer candidate scores can still exhibit outer-maximum optimism when examined after the fact. Effective-rank estimates also depend on matrix construction: raw utilities retain common audit-unit difficulty, row-centred utilities remove level shifts, fixed-reference contrasts depend on the chosen reference, and rank matrices discard utility spacing. These distinctions become acute when the number of candidates exceeds the number of audit units and an empirical candidate-correlation matrix is rank deficient. Finally, retrospective reuse of public outer folds remains distinct from evaluation on an independent external cohort, even when inner fitting is leakage free.

Model-selection behaviour must also be interpreted at chemical-support boundaries. Molecular similarity and scaffold novelty alter prediction difficulty, while activity cliffs show that close structural neighbours can have sharply different properties [9–14,29,30,33]. Rare positive classes pose a separate risk: ROC-AUC can appear favourable even when precision, minority recall or false-negative behaviour is unsuitable for screening. Applicability-domain diagnostics, support-stratified performance, error overlap, uncertainty and conditional coverage therefore define the limits of a selected-model result. Such analyses do not form an independent confirmation of the candidate-expansion finding, and they should not be averaged into a single performance score. Instead, they provide boundary evidence on where apparently strong aggregate selection results may weaken—for dissimilar molecules, novel scaffolds, activity-cliff pairs or scarce toxicity labels.

We therefore conducted a retrospectively specified and subsequently locked, task-stratified audit rather than a model leaderboard. The central question was whether opportunities created by candidate-pool expansion could be realised reproducibly from finite validation information. Ten current seeded scaffold partitions were generated with the locked current split implementation for the 32-candidate registry. Historical unseeded GroupKFold outputs were retained only for data provenance and the split-transition audit. The six-endpoint registry-composition intervention and other analyses without complete ten-seed outputs were designated five-seed secondary analyses. The primary estimand used a K-invariant K = 32 leave-one-seed-out reference; K-dependent cross-fitting and the same-fold maximum were sensitivity and descriptive opportunity quantities.

2 Methods

2.1 Study design and evidence hierarchy

The study retrospectively audited completed experiments in molecular property prediction. A candidate registry was defined as a prespecified, ordered list of complete modelling pipelines, each comprising a molecular representation, preprocessing procedure, learner and fixed or tuned hyperparameters. Endpoints, candidate order, K values, seeds, split logic, selection rules, outcomes and exclusions were held fixed for the analyses reported here. Because the audit specification was reconstructed after the original outer outcomes had been generated, the study was not prospectively preregistered; the term frozen denotes unchanged eligibility and analysis rules during this audit, not an untouched prospective cohort (Figure 1).

Evidence was organized into five levels. Level 1 was the nine-endpoint leave-one-seed-out cross-fitted audit of registered prefixes. Level 2 comprised the descriptive same-fold finite-set opportunity bound, permutation calibration, candidate-composition controls and finite-audit simulations. Level 3 was the matched-size twelve-candidate multiview audit. Level 4 was the six-endpoint expanded registry-composition intervention with component, budget, anchor, normalization, stability and split-mechanism sensitivity analyses. Level 5 comprised prediction-level analyses of reliability and chemical-support boundaries.

These evidence levels address different questions and were not merged into a single leaderboard. Outer folds evaluated frozen inner-selection decisions. The leave-one-seed-out reference is the principal retrospective comparison; the same-fold maximum is an ex post finite-set opportunity bound and neither a deployable selector nor a population truth. The multiview, composition and prediction panels reuse public endpoint definitions and do not constitute independent external validation.

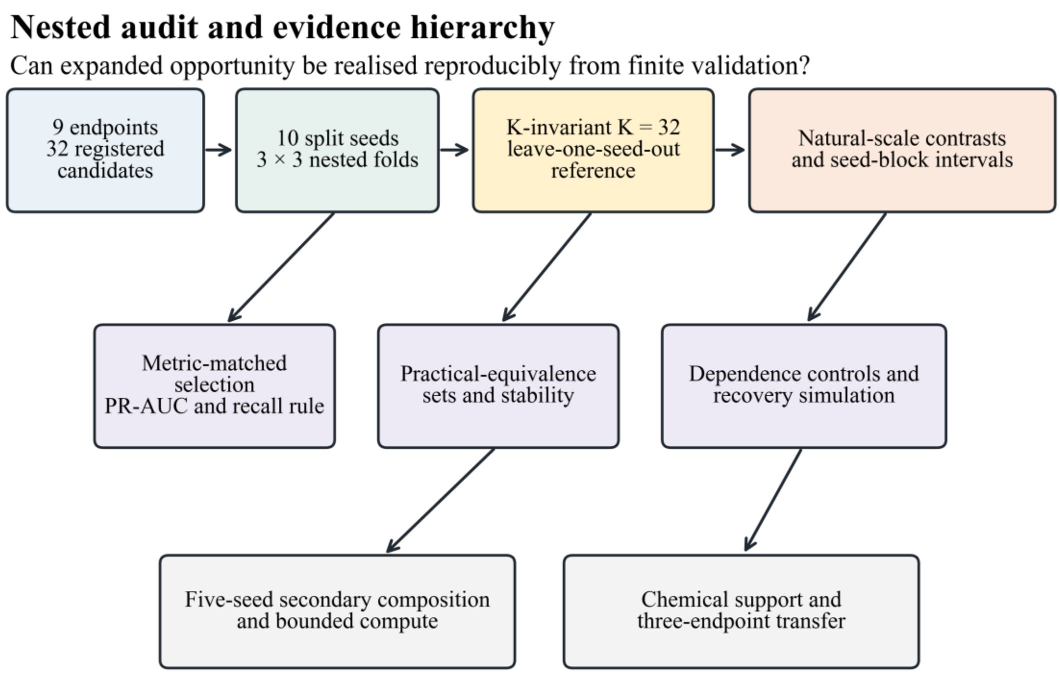

Figure 1. Nested audit and evidence hierarchy. Three inner folds rank registered candidates and three outer folds audit frozen decisions. The lightweight registry uses ten seeded scaffold partitions; secondary composition and reliability panels retain their original prespecified units. K-invariant K = 32 and K-dependent cross-fitted references are separated from the descriptive same-fold opportunity bound, practical-equivalence sets, metric-matched selection, constructed candidate controls, recovery simulation, chemical-support analyses and bounded downstream compute.

2.2 Datasets and molecular standardization

The primary panel comprised ESOL, FreeSolv, Lipophilicity, blood–brain barrier penetration (BBBP), beta-secretase 1 inhibition (BACE), ClinTox, Caco2 Wang, human intestinal absorption (HIA Hou) and P-glycoprotein inhibition (P-gp Broccatelli) [1,2]. ESOL, FreeSolv, Lipophilicity and Caco2 were regression endpoints evaluated by root mean squared error (RMSE). The remaining endpoints were binary classifications evaluated primarily by the area under the receiver operating characteristic curve (ROC-AUC). Precision–recall area under the curve (PR-AUC), Brier score, expected calibration error and minority recall were retained for reliability analyses but did not replace the registered classification utility used for candidate selection.

Molecules were parsed and standardized with RDKit [15]. The largest valid molecular fragment was retained, charge normalization was applied where chemically valid, and canonical SMILES were generated while preserving stereochemical information. These operations preceded feature calculation but were governed by fixed endpoint-level rules rather than by outer performance. Within modelling pipelines, imputation, scaling and any data-dependent transformation were estimated only from the corresponding training fold. For classification endpoints, duplicate structures were merged only when their labels agreed; groups with conflicting labels were removed. For regression endpoints, replicate labels were averaged and the replicate count was retained for the cleaning audit.

Cleaning affected endpoints differently. BBBP was reduced from 2,050 records to 1,955 standardized structures after merging 62 sets of consistent duplicates and excluding 22 conflicting rows and 11 invalid SMILES. ClinTox was reduced from 1,484 to 1,376 structures after excluding 98 conflicting rows and four invalid SMILES. Only 58 ClinTox structures were positive after cleaning, corresponding to 4.2% of the analysis population. This scarcity is reported whenever thresholded toxicity or minority coverage is interpreted. Regression target ranges and units were retained so that raw RMSE effects would not be silently combined across incompatible scales. Dataset-level summaries are provided in Table 1.

Table 1. Primary datasets and endpoint metrics.

Endpoint	n	Class balance / target range	Metric
ESOL	1117	−11.60 to 1.58	RMSE
FreeSolv	642	−25.47 to 3.43	RMSE
Lipophilicity	4200	−1.50 to 4.50	RMSE
BBBP	1955	1494 positive (76.4%)	ROC-AUC
BACE	1512	690 positive (45.6%)	ROC-AUC
ClinTox	1376	58 positive (4.2%)	ROC-AUC
Caco2 Wang	904	−7.76 to −3.51	RMSE

Endpoint	n	Class balance / target range	Metric
HIA Hou	578	500 positive (86.5%)	ROC-AUC
P-gp Broccatelli	1212	647 positive (53.4%)	ROC-AUC

Target units: ESOL, log mol/L; FreeSolv, kcal/mol; Lipophilicity, logD; Caco2 Wang, dataset-provided permeability scale. Classification rows report positive-class n (%).

2.3 Candidate registry and computational exposure

The controlled expansion registry contained 32 candidates using a common Morgan-512 representation. Classification positions 1–4 were logistic-regression variants. Positions 5–12 contained random-forest and extremely randomized-tree variants, positions 13–16 histogram or gradient-boosting variants, positions 17–24 LightGBM variants, positions 25–28 XGBoost variants and positions 29–32 CatBoost variants. The regression registry replaced the first four candidates with ridge and elastic-net models and retained the same tree-family structure. K = 4, 8, 16 and 32 were deterministic prefixes of this fixed order.

Registry order was chosen to add model-family complexity in stages: regularized linear candidates entered first, bagging candidates second and boosting families thereafter. Outer-audit performance was not used to reorder it. Prefix expansion therefore changed both nominal K and family composition, which motivated the separate random-order, random-subset and family-balanced controls. Because the registry contains many candidates that share a fingerprint and closely related learners, it was designed as a controlled stress test of repeated selection rather than as a representative sample of all graph and foundation models.

The evaluation procedure for each candidate was held constant across K. The unique current primary ten-seed matrix comprised 34 560 candidate fits across seeds 11, 23, 37, 53, 71, 83, 97, 113, 127 and 149. The corresponding recorded current-run candidate-fit time was 8 130.16 seconds. Historical five-seed fits and timings are retained only in the supplementary provenance audit and are not added to the current primary exposure.

Exposure units are analysis-specific and are not directly comparable across audit components. Current primary candidate-fit time covers the complete ten-seed lightweight registry; downstream composition time excludes model acquisition, encoder pretraining and cached embedding extraction. Historical timings remain in the provenance table and are not combined with current totals.

Table 2. Audit components and recorded computational exposure.

Audit component	Registry	Evaluation design	Recorded exposure
Controlled prefix audit	32 Morgan candidates	9 endpoints; K = 4, 8, 16, 32; 10 seeds; 3 outer × 3 inner folds	34 560 candidate fits; candidate-fitting time 8 130.16 s
Calibration and resampling controls	Stored 32-candidate results	5 000 permutations; 100 resamples per mode/K/seed	No additional model fitting
Matched multiview audit (secondary)	12 multiview candidates	9 endpoints; 5 seeds; all C(12,3) registered subsets	6 480 candidate fits
Split-regime transfer audit (secondary)	32 Morgan candidates	3 endpoints; seeded scaffold and Tanimoto-component splits	5 760 candidate fits
Registry-composition intervention (secondary)	Three locked candidate registries	6 endpoints; K = 4, 8, 16, 32; composition and stability controls	1 080 outer units; downstream fitting/prediction time 64 616.35 s

2.4 Repeated nested scaffold evaluation

Ten current seeded scaffold partitions were generated independently for every classification and regression endpoint using seeds 11, 23, 37, 53, 71, 83, 97, 113, 127 and 149. Each partition contained three outer scaffold folds and three inner scaffold folds within every outer-training partition. Bemis–Murcko scaffold groups were kept intact; group allocation used seed-dependent random tie-breaking while balancing sample counts, and no model performance informed allocation. The split manifest records sample and scaffold counts, target summaries, split hashes and the absence of cross-fold scaffold overlap.

The validation-best selector ranked candidates by mean inner utility, where utility was ROC-AUC for classification and negative RMSE for regression. Ties were resolved deterministically by registry order. The chosen candidate was refitted on the complete outer training partition and evaluated once on the held-out outer fold. Fixed-single, one-standarderror and risk-adjusted selectors were retained as sensitivity analyses, but validation-best defined the primary selection-loss contrast. Candidate failures remained in the audit trail and did not trigger outer-informed replacement.

Because outer folds have overlapping training sets, they are paired audit units rather than independent biological replications. For every primary endpoint, fold effects were first averaged within each of ten distinct split seeds before split-seed resampling. The three folds within a seed do not constitute three independent replications. Analyses with only seeds 11, 23, 37, 53 and 71 are explicitly labelled five-seed secondary analyses.

2.5 Cross-fitted selection gaps and ranking estimands

Formal mathematical definitions are concentrated in Section 2.15; this section explains the scientific interpretation, direction and aggregation level of each estimand. The primary selection gap compares the validation-selected candidate with a leave-one-seed-out cross-fitted reference whose identity is determined without the held-out seed. The same-fold finite-set maximum is reported separately as an ex post opportunity bound. Its corresponding gap is an ROC-AUC difference for classification and an RMSE increase for regression because regression utility is negative RMSE, but it cannot be interpreted as deployable regret or a population generalization loss. Range-normalized quantities remain sensitivity estimands rather than universal chemical effects.

CAHit@3 asks whether the observed finite-audit best appears in the validation top three after correction for its random-order recovery probability, whereas normalized MRR measures its complete validation position relative to the random-order expectation. NDCG, Spearman correlation, Kendall correlation and validation-rank percentile were retained as complementary ranking summaries. The permutation and graded signal-injection controls calibrated metric zero points and signal recovery; they did not alter empirical candidate selections.

For the primary lightweight-registry estimands, unit-level metrics were averaged over three outer folds within each split seed and then summarized over ten seed blocks within endpoint. Five-seed secondary panels retained their original seed blocks and are labelled as such. Endpoint-level medians, interquartile ranges and ranges were used for cross-endpoint descriptive summaries; classification and regression were not pooled because ROC-AUC differences and RMSE increases are not exchangeable units.

2.6 Effective-diversity estimation

For the ten-seed primary effective-diversity audit, each endpoint and K produced a 30 × K outer-utility matrix and a 90 × K inner-utility matrix from ten split seeds, three outer folds and three inner folds per outer-training partition. Four prespecified transformations were analysed: raw utilities; row-centred utilities; fixed-reference-relative utilities; and within-unit ranks. The ten-seed outer matrix is shown in Figure 2 and Table S6; the corresponding inner matrix is reported in Table S6. Five-seed composition and multiview matrices remain separate secondary analyses and are not mixed with these estimates.

Effective diversity was derived from candidate-utility matrices rather than candidate labels or model-family counts. Outer matrices characterize behavioural similarity on held-out scaffold groups, whereas inner matrices assess whether redundancy is already visible during selection. Candidate columns with numerical variance at or below 10^{-12} were excluded before correlation estimation and their post-transformation column counts were recorded. Fixed-reference-relative matrices subtract candidate 1 from every other candidate and therefore contain K − 1 columns.

With n rows, a centred empirical K × K correlation matrix has rank at most n − 1 and can be unstable even when K < n. We therefore report both the empirical spectrum and Ledoit–Wolf shrinkage after column standardization. The shrinkage covariance was converted to a correlation matrix before eigenanalysis; this stabilizes the spectrum but does not create independent audit information.

Let $\lambda_1,\ldots,\lambda_K$ be the non-negative eigenvalues of R and $p_i = \lambda_i/\Sigma_j \lambda_j$. Spectral-entropy effective rank was $r_{entropy} = \exp(-\Sigma_i p_i \log p_i)$. Participation-ratio effective rank was $r_{PR} = (\Sigma_i \lambda_i)^2/\Sigma_i \lambda_i^2$, equivalently $1/\Sigma_i p_i^2$. Both equal one for a perfectly redundant rank-one spectrum and K when all eigenvalues are equal. Spectral entropy is more sensitive to small eigenvalue mass, whereas the participation ratio places greater weight on dominant directions. Median off-diagonal correlation was reported as a directly interpretable complement.

Sensitivity of endpoint-specific effective diversity was assessed by omitting each of the ten split seeds, each outer-fold label and, for inner matrices, each inner-fold label, followed by complete recomputation of shrinkage correlations and effective ranks. Fixed-reference-relative estimates were also repeated with three additional registry-prespecified reference candidates; no reference was selected from outer performance. Across-endpoint summaries are medians, IQRs and ranges, without a population-level confidence interval because the nine endpoints were not treated as a random sample from a molecular-task population (Tables S6–S7).

2.7 Candidate-composition controls

Prefix composition could confound a K effect because each increase added specific model families. Three frozen controls addressed this issue. Random-order controls permuted registry order before taking prefixes. Random-subset controls sampled K candidates from the complete registry without requiring a prefix. Family-balanced controls sampled candidates to maintain representation of available learner families. Each control used 100 prespecified seeds per endpoint and K.

The subset or order seed was a resampling device, not an independent endpoint. Results were first summarized within endpoint, and endpoint remained the cross-task inferential unit. At K = 32 all three modes necessarily reproduce the complete pool; convergence at that point is therefore a design property rather than independent corroboration. The controls test whether increasing normalized selection loss was peculiar to one early registry composition, not whether all conceivable candidate distributions would show the same trend.

2.8 Leave-one-seed-out references and completion-gap decomposition

For outer unit u = (s,f), the validation-selected candidate ĵ_u(K), the K-invariant full-registry reference $j_{\mathrm{ref}}^{\mathrm{inv}}(-s)$, the K-dependent within-prefix reference $j_{\mathrm{ref}}^{\mathrm{dep}}(-s,K)$, the K-invariant completion gap $G_{\mathrm{inv}}(u,K)$, the K-dependent selection gap $G_{\mathrm{dep}}(u,K)$, the same-fold finite-set opportunity gap $G_{\mathrm{same}}(u,K)$ and the availability component $G_{\mathrm{avail}}(u,K)$ are defined formally in Equations (6)–(13). The exact identity $G_{\mathrm{inv}}(u,K) = G_{\mathrm{avail}}(u,K) + G_{\mathrm{dep}}(u,K)$ separates candidate availability from within-prefix selection. The seed-level mean $\bar{G}_{\mathrm{inv},e,s}(K)$ first averages the three outer folds, and the endpoint contrast $\Delta_{\mathrm{inv}}(e)$ then averages paired differences over the ten seed blocks; a negative K = 32 minus K = 4 contrast indicates a smaller full-registry completion gap at K = 32, whereas a positive contrast indicates a smaller gap at K = 4.

Reference identities were selected without the held-out seed and registry order resolved ties. The same-fold maximum is descriptive rather than deployable. Seed isolation reduces same-unit circularity but reuses the same public endpoints, candidate registry and split generator; the ten seed blocks are sensitivity replicates, not independent studies or external validation.

2.9 Finite-audit winner-optimism simulation

A known-truth simulation isolated the mechanical effect of maximizing noisy audit estimates. In the equal-truth scenario, all candidates had identical true standardized utility. Audit errors followed a multivariate Gaussian distribution with equicorrelations of 0, 0.5 or 0.9. Effective audit sample sizes were 25, 50, 100 or 200, candidate counts were $K = 4, 8, 16$ or 32 and 30,000 Monte Carlo replicates were generated for every configuration. A weak-gradient scenario was retained as a secondary sensitivity analysis.

Winner optimism was defined as the maximum observed audit utility minus the true utility of the candidate attaining that maximum, expressed in simulation standard-deviation units. Because candidate truth was equal in the primary simulation, any positive observed maximum was produced entirely by finite estimation noise. The simulation was not used to subtract a bias correction from empirical endpoints. Its purpose was to demonstrate that an outer maximum can be optimistic even when the outer data were not used for inner selection and even when candidate estimates are strongly correlated.

2.10 Matched-size multiview stress test

The multiview registry crossed four representations (Morgan-512, MACCS, RDKit2D and their concatenation) with three learners. The matched-size analysis exhaustively enumerated all $C(12,3) = 220$ three-candidate subsets without retraining, using stored candidate-level inner and outer utilities. These overlapping subsets are composition-sensitivity contrasts within one registry, not independent experiments; the endpoint remained the interpretation unit and no subset-level P values were calculated. Subsets were assigned to mutually exclusive classes: Morgan-only reference, single-representation only, single-learner only, representation-balanced without concatenation, representation-balanced with concatenation, or mixed unbalanced. We report medians, IQRs, distribution ranges and the within-endpoint proportion of subsets that outperformed Morgan-only.

Within-learner analyses selected among four representations for one learner and compared the result with that learner's Morgan candidate. The full K = 12 versus Morgan K = 3 contrast is termed the multiview-pool gain because it combines representation breadth and additional candidates. Only comparisons at the same K are termed representation-composition effects.

For every paired unit we retained the endpoint, split seed, outer fold, registered candidates, selected candidate and paired inner and outer utilities. Deterministic split hashes and SHA-256 source hashes were recorded in the machine-readable supplement. Prediction-level quantities were calculated only for registries with traceable candidate-level prediction exports.

2.11 Limited representation-baseline analysis

A separate six-endpoint panel included RDKit random forest (RDKit-RF), a graph convolutional network (GCN), a ChemBERTa frozen-embedding probe and a MoLFormer frozen-embedding probe. RDKit-RF used Morgan fingerprints and 120 trees. The GCN contained two graph-convolution layers with 32 hidden units each, global mean pooling and five fixed training epochs. ChemBERTa and MoLFormer encoders were frozen and followed by logistic or ridge probes. These configurations provide representation diversity but are not exhaustive or equivalently tuned modern baselines.

Because search budgets differed across candidates, the limited four-model panel was restricted to selector behaviour and error correlation under shared splits. D-MPNN was not included in that prediction-level panel; a separately locked one-epoch D-MPNN candidate was included only in the expanded registry-composition intervention. TabPFN was not included because a complete shared-split audit was unavailable.

Per-molecule outer predictions were available for the rerun 32-candidate regression audit and for the completed four-model representation panel. Prediction correlations, high-error

Jaccard overlap, Ledoit–Wolf entropy rank and participation-ratio rank were calculated only where the requisite candidate-level predictions were traceable. Prediction-level diversity is therefore reported as a boundary analysis and is not treated as equivalent to utility-pattern diversity.

2.12 Reliability and chemical-boundary analyses

Classification uncertainty was evaluated with split conformal, label-conditional conformal and Mondrian label-and-similarity conformal prediction at 80%, 90% and 95% nominal coverage. Minority-class coverage and prediction-set size were primary reliability outputs because overall marginal coverage can conceal undercoverage of rare positives. Regression intervals used conformalized quantile regression (CQR). Coverage and interval width were reported separately by endpoint because interval widths inherit endpoint units [26–28,33,36].

Ensemble uncertainty was assessed by its association with absolute error, risk-coverage behaviour and high-error enrichment. Test molecules were stratified by maximum training-set Morgan Tanimoto similarity into low (<0.5), intermediate (0.5 to <0.7) and high (at least 0.7) support. We compared selected and cross-fitted-reference performance, candidate error overlap, model disagreement, uncertainty-error association and false-negative enrichment within each stratum. Exact held-out Murcko scaffolds were unseen by construction; related versus novel scaffolds were therefore defined by a prespecified maximum scaffold-fingerprint similarity threshold of 0.5. This derived relatedness definition is reported explicitly.

ClinTox was treated as a prespecified negative reliability example because only 58 positive standardized structures were available. ROC-AUC, PR-AUC, minority recall, false-negative rate, label-conditional conformal coverage and mean prediction-set size were reported as distinct quantities. Coverage addresses prediction-set validity, whereas recall and false-negative rate address screening safety; none of these quantities establishes deployment-level toxicology performance.

These analyses were included in the main text only to bound interpretation. Supplementary methods and extended audit results are provided in Additional file 1; machine-readable source and result tables, including dataset access and licensing records, are provided in Additional file 2; and Supplementary Figures S1-S21 are provided in Additional file 3.

2.13 Split-regime transfer audit

ClinTox, BACE and ESOL represented rare-class classification, regular classification and regression, respectively. The locked 32-candidate Morgan-512 registry, candidate order, K = 4, 8, 16 and 32 prefixes, seeds 11, 23, 37, 53 and 71, and three outer folds, each with three inner folds, were retained. Scaffold results reused the locked source-of-record outputs. For the new structure-separated rerun, molecular pairs with Morgan-512 Tanimoto similarity of at least 0.70 were connected, and each connected component was kept intact during outer and inner allocation. The threshold and allocation rule were fixed before formal model fitting after splitter feasibility checks; model performance did not inform grouping.

A seeded greedy allocator balanced sample counts and, for classification, both class counts across intact components. Every formal inner and outer split was required to retain both classes where applicable, keep components disjoint and have maximum cross-fold Tanimoto below 0.70; random fallback was prohibited. The transfer audit added 5,760 candidate fits. Transfer was assessed for the direction of the CAHit@3 change from K = 4 to K = 32, leave-one-seed-out cross-fitted K = 32 minus K = 4 selection gaps and Ledoit–Wolf effective ranks under the four prespecified matrix transformations. This was a robustness analysis on three selected endpoints, not a new confirmatory population sample.

2.14 Expanded equal-size registry-composition intervention

Endpoint and registry design

Six endpoints were fixed from task coverage before expanded outcomes were read: ClinTox, BACE and BBBP for classification, and ESOL, Lipophilicity and Caco2 Wang for regression. Homogeneous Morgan, classical multiview and modern-augmented exact-prefix registries were evaluated at K = 4, 8, 16 and 32 with seeds 11, 23, 37, 53 and 71 and identical nested partitions with three outer folds and three inner folds per outer-training set. The shared Morgan linear candidate occupied the first locked position; candidate configuration and order did not change across eligible prefixes.

Component ablations

Fixed-K ablations compared classical multiview, classical plus ChemBERTa, classical plus MoLFormer, classical plus D-MPNN and the full modern-augmented registry at K = 16 and 32. Each addition replaced a prespecified classical candidate so that K remained exact. The separately locked one-epoch D-MPNN belonged only to this expanded intervention.

Equal-downstream-budget analysis

Each locked prefix was truncated at the endpoint- and K-specific median downstream time of the classical multiview registry; outer performance never determined retention. Recorded time comprised inner and outer downstream fit/predict time and excluded model acquisition, encoder pretraining and cached embedding extraction. This analysis therefore assesses bounded downstream compute rather than end-to-end efficiency.

Anchor, normalization, selection-stability and split-regime sensitivity

Anchor sensitivity used the shared Morgan linear model, a fixed Morgan random forest and the predefined registry-median candidate. Raw gains, endpoint-MAD normalization and paired homogeneous-audit-best normalization were reported. Candidate, representation and learner-family frequencies, normalized selection entropy, modal proportion, leave-one-seed-out agreement and adjacent-fold switches quantified stability. ClinTox, BACE and ESOL were evaluated across all three registries under seeded scaffold and Tanimoto-component splits with registries, seeds, folds, rules and metrics held fixed.

2.15 Mathematical estimands and matrix-relative diversity

Let u = (s, f) index an outer audit unit, V(u,j) the mean inner-validation utility and A(u,j) the outer-audit utility. For the primary audit S_{main} = 10 and F = 3; for explicitly labelled secondary analyses $S_{\text{secondary}}$ = 5. The eligible prefix $\mathcal{C}_K$, candidate order and tie rules were fixed before the reported contrasts were calculated. The notation table and equations define the validation-selected candidate, K-invariant and K-dependent cross-fitted references, same-fold finite-set best, G_{inv}, G_{dep}, G_{same}, G_{avail}, CAHit@3, normalized MRR, entropy effective rank and normalized selection entropy.

Table 3. Mathematical notation used in the audit estimands.

Symbol	Definition
S_{main}	number of primary split seeds (10)
$S_{\text{secondary}}$	number of seeds in explicitly labelled secondary analyses (5)
s	split seed
F	number of outer folds per seed (3)
f	outer fold
$u = (s, f)$	outer audit unit
U_{-s}	outer audit units from seeds other than held-out seed s
N_{-s}	number of units in U_-s; N_-s = |U_-s|
j	candidate
K	candidate count
$\mathcal{C}_K$	eligible registered candidate prefix at size K
$V(u, j)$	inner-validation utility
$A(u, j)$	outer-audit utility (negative RMSE direction for regression)
τ	practical-equivalence reporting tolerance

Table 3 (continued). Mathematical notation used in the audit estimands.

Symbol	Definition
$\varepsilon_{\mathrm{num}}$	numerical-stability constant, fixed at 10^-12
$\hat{j}_u(K)$	validation-selected candidate at size K
$j_{\mathrm{ref}}^{\mathrm{inv}}(-s)$	K-invariant full-registry reference selected without seed s
$j_{\mathrm{ref}}^{\mathrm{dep}}(-s,K)$	K-dependent eligible-prefix reference selected without seed s
$G_{\mathrm{inv}}(u,K)$	K-invariant full-registry completion gap
$G_{\mathrm{dep}}(u,K)$	K-dependent within-prefix selection gap
$G_{\mathrm{same}}(u,K)$	same-fold finite-set opportunity gap
$G_{\mathrm{avail}}(u,K)$	availability component; G_{inv} = G_{avail} + G_{dep}
$\bar{G}_{\mathrm{inv},e,s}(K)$	seed-level mean of G_{inv} across the three outer folds
$\Delta_{\mathrm{inv}}(e)$	endpoint-level K=32 minus K=4 contrast averaged over seed blocks
$\mathbf{S}_{\mathrm{cov}}$	sample covariance matrix in Ledoit-Wolf shrinkage
λ_i	non-negative shrinkage-correlation eigenvalue
p_i	unit-sum eigenvalue proportion
π_j	selection proportion for candidate j

Selection and ranking estimands

Here r_u is the inner-validation rank of the outer finite-audit best candidate, q is a rank cutoff (q = 3), I is the indicator function and H_K is the Kth harmonic number.

Equations (1)–(5) define candidate selection, finite-audit selection loss and chance-adjusted ranking. Equations (6)–(13) define the validation-selected candidate, K-invariant and K-dependent references, the four gaps, their availability decomposition and the endpoint-level Δ_{inv} contrast. Equations (14)–(17) define matrix transformations and effective ranks; Equations (18)–(21) define composition contrasts, normalized gains and selection entropy.

$$\hat{j}_u(K) = \arg\max_{j \in \mathcal{C}_K} V(u,j) \tag{1}$$

$$j_{\mathrm{best},u}(K) = \arg\max_{j \in \mathcal{C}_K} A(u,j) \tag{2}$$

$$L_u(K) = A(u, j_{\mathrm{best},u}(K)) - A(u, \hat{j}_u(K)); \tilde{L}_u(K) = \frac{L_u(K)}{\max_{j \in \mathcal{C}_K} A(u,j) - \min_{j \in \mathcal{C}_K} A(u,j) + \varepsilon_{\mathrm{num}}} \tag{3}$$

$$CAHit_{q,u} = \frac{I(r_u \le q) - \frac{q}{K}}{1 - \frac{q}{K}}, q = 3 \;\; ; \;\; MR\mathbf{R}_u = \frac{1}{r_u} \tag{4}$$

$$E_{0(MRR)} = \frac{H_K}{K} \;\; ; \;\; MR\widetilde{\mathbf{R}}_u = \frac{MR\mathbf{R}_u - \frac{H_K}{K}}{1 - \frac{H_K}{K}} \tag{5}$$

Completion-gap estimands and decomposition

Let U_{-s} denote the outer audit units from seeds other than held-out seed s and define N_{-s} = $|U_{-s}|$. The reference-training set contains all outer units from seeds other than held-out seed s. Primary references use the other nine of S_{main} = 10 seeds; five-seed secondary analyses use the other four only within their own scope. Equations (6)–(13) define the candidate identities, G_{inv}, G_{dep}, G_{same}, G_{avail}, the exact decomposition and $\Delta_{\mathrm{inv}}(e)$. A negative $\Delta_{\mathrm{inv}}(e)$ means a smaller full-registry completion gap at K = 32; a positive value means a smaller gap at K = 4.

$$\hat{\mathrm{j}}_{\mathrm{u}}(\mathrm{K}) \equiv \mathrm{j}_{\mathrm{u}}^{\mathrm{K}} = \arg\max_{\mathrm{j} \in \mathrm{C}_{\mathrm{K}}} \mathrm{V}(\mathrm{u}, \mathrm{j}) \tag{6}$$

$$j_{\mathrm{ref}}^{\mathrm{inv}}(-s) = \arg\max_{j \in \mathcal{C}_{32}} \frac{1}{N_{-s}} \sum\nolimits_{u \in U_{-s}} A(u,j) \tag{7}$$

$$j_{\mathrm{ref}}^{\mathrm{dep}}(-s, K) = \arg\max_{j \in \mathcal{C}_K} \frac{1}{N_{-s}} \sum\nolimits_{u \in U_{-s}} A(u,j) \tag{8}$$

$$\begin{gathered} \mathrm{G}_{\mathrm{inv}}(\mathrm{u}, \mathrm{K}) = \mathrm{A}(\mathrm{u}, \mathrm{j}_{\mathrm{ref}}^{\mathrm{inv}}(-\mathrm{s})) - \mathrm{A}(\mathrm{u}, \hat{\mathrm{j}}_{\mathrm{u}}(\mathrm{K})) \\ \mathrm{G}_{\mathrm{dep}}(\mathrm{u}, \mathrm{K}) = \mathrm{A}(\mathrm{u}, \mathrm{j}_{\mathrm{ref}}^{\mathrm{dep}}(-\mathrm{s}, \mathrm{K})) - \mathrm{A}(\mathrm{u}, \hat{\mathrm{j}}_{\mathrm{u}}(\mathrm{K})) \end{gathered} \tag{9}$$

$$\mathrm{G}_{\mathrm{same}}(\mathrm{u}, \mathrm{K}) = \max_{\mathrm{j} \in \mathrm{C}_{\mathrm{K}}} \mathrm{A}(\mathrm{u}, \mathrm{j}) - \mathrm{A}(\mathrm{u}, \hat{\mathrm{j}}_{\mathrm{u}}(\mathrm{K})) \tag{10}$$

$$\begin{gathered} \mathrm{G}_{\mathrm{avail}}(\mathrm{u}, \mathrm{K}) = \mathrm{A}(\mathrm{u}, \mathrm{j}_{\mathrm{ref}}^{\mathrm{inv}}(-\mathrm{s})) - \mathrm{A}(\mathrm{u}, \mathrm{j}_{\mathrm{ref}}^{\mathrm{dep}}(-\mathrm{s}, \mathrm{K})) \\ \mathrm{G}_{\mathrm{inv}}(\mathrm{u}, \mathrm{K}) = \mathrm{G}_{\mathrm{avail}}(\mathrm{u}, \mathrm{K}) + \mathrm{G}_{\mathrm{dep}}(\mathrm{u}, \mathrm{K}) \end{gathered} \tag{11}$$

$$\bar{G}_{\mathrm{inv},e,s}(K) = \frac{1}{F} \sum_{f=1}^{F} G_{\mathrm{inv}_e}((s,f), K) \tag{12}$$

$$\Delta_{\mathrm{inv}}(e) = \frac{1}{S_{\mathrm{main}}} \sum_{s=1}^{S_{\mathrm{main}}} [\, \bar{G}_{\mathrm{inv},e,s}(32) - \bar{G}_{\mathrm{inv},e,s}(4)] \tag{13}$$

Matrix transformations and effective rank

X is the outer-utility matrix; $\mathbf{S}_{\mathrm{cov}}$ in the Ledoit–Wolf expression denotes sample covariance and must not be confused with $S_{\mathrm{main}} = 10$ or $S_{\mathrm{secondary}} = 5$. T is the scaled-identity target, λ_i are the non-negative eigenvalues of the shrinkage correlation matrix and p_i their unit-sum proportions. Equations (14)–(17) define the transformations and effective ranks.

$$\mathbf{X}_{raw,u,j} = A_{u,j} \;\; ; \;\; \mathbf{X}_{ctr,u,j} = A_{u,j} - \left(\frac{1}{K}\right) \sum_l A_{u,l} \tag{14}$$

$$\mathbf{X}_{ref,u,j} = A_{u,j} - A_{u,j_0} \;\; ; \;\; \mathbf{X}_{rank,u,j} = rank_{j(A_{u,j})} \tag{15}$$

$$\hat{\Sigma}_{LW} = (1-\alpha) S_{cov} + \alpha T \;\; ; \;\; p_i = \frac{\lambda_i}{\sum_l \lambda_l} \tag{16}$$

$$r_{ent} = \exp[-\textstyle\sum_i p_i \ln(p_i)] \;\; ; \;\; r_{PR} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2} = \frac{1}{\sum_i p_i^2} \tag{17}$$

Composition and normalization

The label hom denotes the homogeneous Morgan pool; complete downstream-efficiency and Pareto definitions are provided in Supplementary Methods. Equations (18)–(21) define the composition contrasts, normalized gains and normalized selection entropy.

$$\begin{gathered} G_{best,u,p,K} = \max_{j \in \mathcal{C}_{p,K}} A_{u,j} - A_{u,a} \\ G_{sel,u,p,K} = A_{u,\hat{j}_{u,p,K}} - A_{u,a} \end{gathered} \tag{18}$$

$$\begin{gathered} \tilde{G}_{sel,e,p,K} = \frac{1}{SF} \sum_{s,f} \frac{G_{sel,(s,f),p,K}}{G_{best,(s,f),hom,K}} \\ d_{rel,e,p,K} = \frac{r_{ent}(\mathbf{x}_{e,p,K})}{K} \end{gathered} \tag{19}$$

$$\tilde{L}_{CF,e,p,K} = \frac{1}{SF} \sum_{s,f} \frac{\left[A_{(s,f),j_{ref-s}} - A_{(s,f),\hat{j}_{(s,f)}}\right]}{G_{best,(s,f),hom,K}} \tag{20}$$

Selection stability

Here π_j is the proportion of repeated outer audit units in which candidate j is selected; it is distinct from the rank cutoff q in the CAHit@q definition. Equation (21) defines normalized selection entropy.

$$H_{sel} = -\sum_j \pi_j \ln(\pi_j) \quad ; \quad H_{sel,norm} = \frac{H_{sel}}{\ln(K)} \tag{21}$$

We used the numerical-stability constant $\varepsilon_{\mathrm{num}} = 10^{-12}$. Ledoit–Wolf shrinkage used the scaled-identity target and the analytic coefficient implemented in scikit-learn. The shrinkage covariance was rescaled to a correlation matrix R before eigenanalysis. Correlation eigenvalues below zero only because of floating-point error were clipped to zero, with the convention $0\log 0 := 0$. Paired normalized gains and cross-fitted gaps were computed within paired outer units, averaged over the three outer folds within each seed, and then summarized within endpoint using seed as the block; units with an absolute denominator at or below $\varepsilon_{\mathrm{num}}$ were reported as missing.

2.16 Statistical uncertainty and scope of inference

For endpoint-specific K = 32 minus K = 4 contrasts in the lightweight registry, fold differences were averaged within each of ten split seeds and seed blocks were resampled 10,000 times. Classification used seeded stratified scaffold-group folds; regression used seeded scaffold-group-balanced folds for all ten current runs. The legacy unseeded regression GroupKFold outputs were retained only for a split-transition audit and were not mixed into the primary estimand. Resamples reduce Monte Carlo error but do not increase the ten-seed information base; the reported 95% intervals are descriptive split-seed sensitivity intervals, not population confidence intervals for unseen molecular endpoints. Classification ROC-AUC and regression RMSE effects were not pooled, and no endpoint-aggregation P value or confirmatory multiplicity family was defined.

Endpoint–pool–K cells are not independent. Fold effects were averaged within seed, seed-block uncertainty was retained and direction counts were interpreted descriptively. Component and budget comparisons were paired within endpoint, seed and fold. Matched-size results summarize 220 overlapping registered subsets rather than independent experiments; support and scaffold analyses retain classification and regression on separate numerical axes.

2.17 Practical equivalence and near-equivalent sets

Retrospectively locked practical-equivalence tolerance τ grids were 0.005, 0.010 and 0.020 ROC-AUC for classification and 0.025, 0.050 and 0.100 RMSE for regression. The middle values were working thresholds, not clinical or pharmacological minimum important differences. We reported τ-regret, τ-success, validation- and cross-fitted near-equivalent set sizes, set Jaccard overlap and whether the selected candidate belonged to the cross-fitted set.

2.18 PR-AUC and minority-recall-constrained selection

The five classification endpoints were refitted on the frozen ten-seed nested splits to record inner ROC-AUC, PR-AUC and operating-point metrics. In each inner fold, the minority label was the less frequent class in that fold's training partition. All distinct ROC thresholds from the inner validation predictions were evaluated. If one or more thresholds attained minority recall ≥ 0.80, the threshold maximizing majority recall was selected; ties were resolved by higher minority recall, higher balanced recall and then proximity to 0.5. If none attained 0.80, the same lexicographic rule was applied to all finite-recall thresholds and the failure of threshold feasibility was recorded. Candidate-specific thresholds were aggregated as the median of the three inner-fold thresholds, and the minority label as their mode, without post-hoc probability calibration. Candidate selection first retained candidates whose mean inner minority recall was ≥ 0.80; if none qualified, the full eligible prefix was retained. It then maximized mean inner majority recall, with ties resolved by mean inner minority recall, mean inner PR-AUC and registry order. The candidate identity, aggregated threshold and minority label were frozen before one-time outer-fold evaluation. We report threshold feasibility, candidate eligibility, outer target satisfaction, minority and majority recall, minority miss rate, PR-AUC and ROC-AUC.

2.19 Constructed controls, order sensitivity and recovery simulation

Exact duplicates copied registered utility vectors; near-duplicates combined 95% of an anchor with 5% of a donor utility vector. Weak and complementary-strong additions were chosen using only non-held-seed mean utility and utility-pattern correlation. Registry, cross-fitted best-first, worst-first, cost-first and 100 deterministic random orders were evaluated. Recovery simulations treated outer candidate utilities as fixed truth and added either an empirical candidate-correlated heteroskedastic residual process or an independent homoskedastic comparator at 0.5, 1 and 2 times the estimated noise scale, with 2,000 replicates per endpoint, K and scenario.

3 Results

3.1 Effective diversity depended on matrix construction

In the ten-seed primary outer effective-diversity audit, the K = 32 endpoint medians of Ledoit–Wolf entropy rank were 2.36 for raw utilities, 19.25 after row centring, 5.56 for fixed-reference-relative utilities and 24.33 for within-unit ranks. These matrices retain common audit-unit difficulty, remove common level shifts, express contrasts to a prespecified reference or preserve ordering without utility spacing, respectively. The corresponding 30 × 32 outer and 90 × 32 inner matrices are reported without combining five-seed secondary composition analyses (Figure 2; Tables S6–S7).

Matrix level, estimator, leave-one-seed, leave-one-fold and reference sensitivities changed magnitudes but preserved the distinction between nominal K and matrix-dependent effective rank. Full IQRs, ranges, participation-ratio ranks, shrinkage coefficients, correlation summaries and inner-matrix results are in Table S6; leave-one-unit and predefined-reference results are in Table S7.

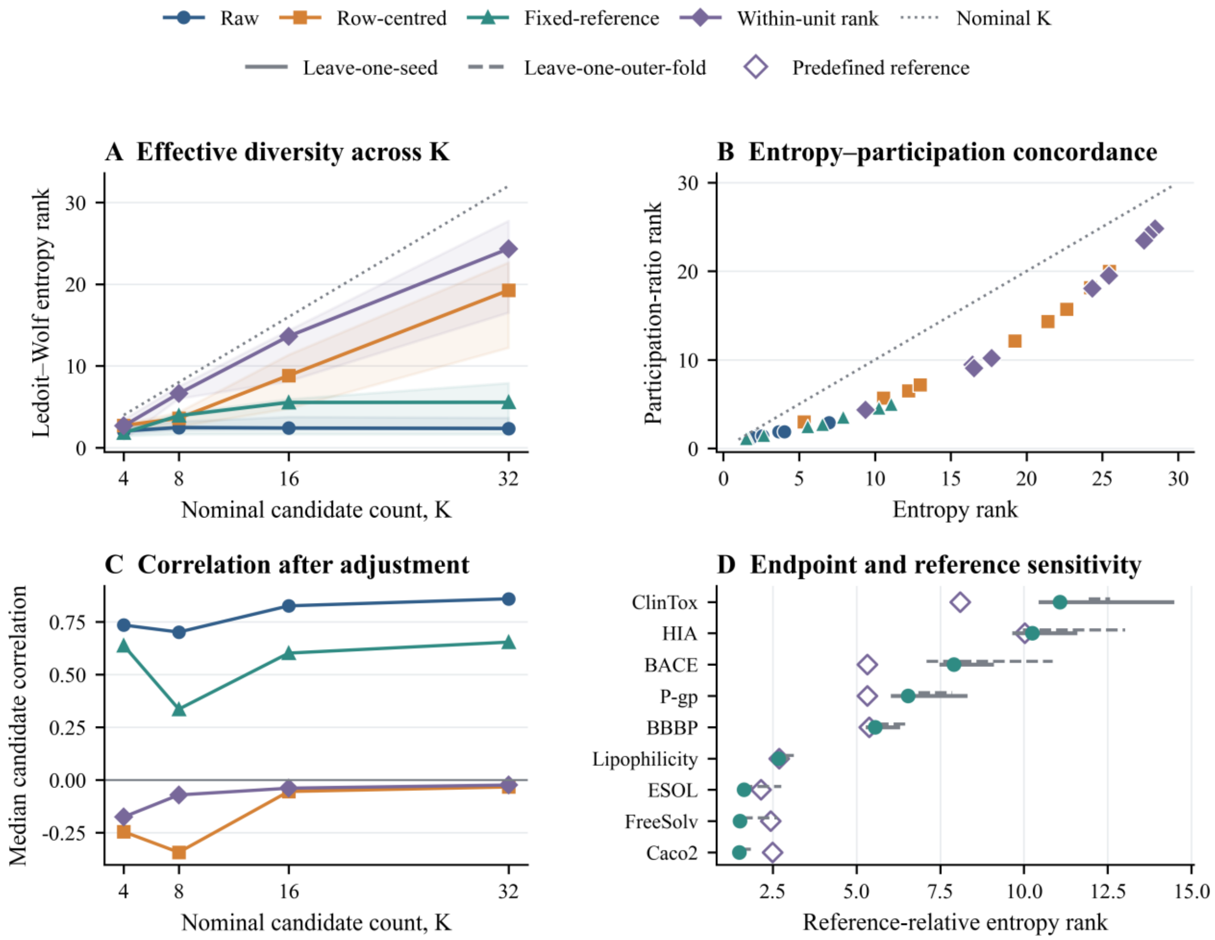

Figure 2. Ten-seed matrix-dependent candidate diversity. (A) Ledoit–Wolf entropy rank across raw, row-centred, fixed-reference-relative and within-unit-rank outer-utility matrices; lines are endpoint medians, ribbons are endpoint IQRs and the grey dotted line is nominal K. (B) Entropy and participation-ratio concordance at K = 32 with a 45° reference. (C) Median candidate correlation after adjustment. (D) Fixed-reference-relative endpoint estimates at K = 32; solid and dashed ranges omit one seed and one outer-fold label, respectively, filled teal points are complete ten-seed estimates and hollow diamonds use a predefined linear reference. Each outer matrix contains 30 seed–fold rows; corresponding 90-row inner-matrix results are reported in Tables S6–S7.

3.2 Chance-adjusted ranking fidelity declined with K

The permutation negative control was centred near zero after chance adjustment. As the injected validation–audit signal increased, CAHit@3 and normalized MRR increased, whereas fixed-range selection loss decreased. Under the specified permutation and signal-injection conditions, these metrics therefore showed the expected zero-point and signal-recovery behaviour, but this diagnostic does not establish a biochemical mechanism (Figure 3A–B).

3.3 Expansion increased observed opportunity more than selected gain

For classification, mean observed audit-best gain increased from 0.0028 to 0.0311 ROC-AUC between $K = 4$ and $K = 32$; selected-model gain increased from −0.0014 to 0.0173. For regression, observed audit-best gain increased from 0.7619 to 0.8078 RMSE units; selected-model gain changed from 0.7442 to 0.7378; classification and regression scales were not pooled.

Across both task strata, expansion increased observed opportunity faster than realised gain. Near-zero audit-best denominators made realisation ratios unstable, so they remain secondary machine-readable quantities (Figure 4A–B; Table S8).

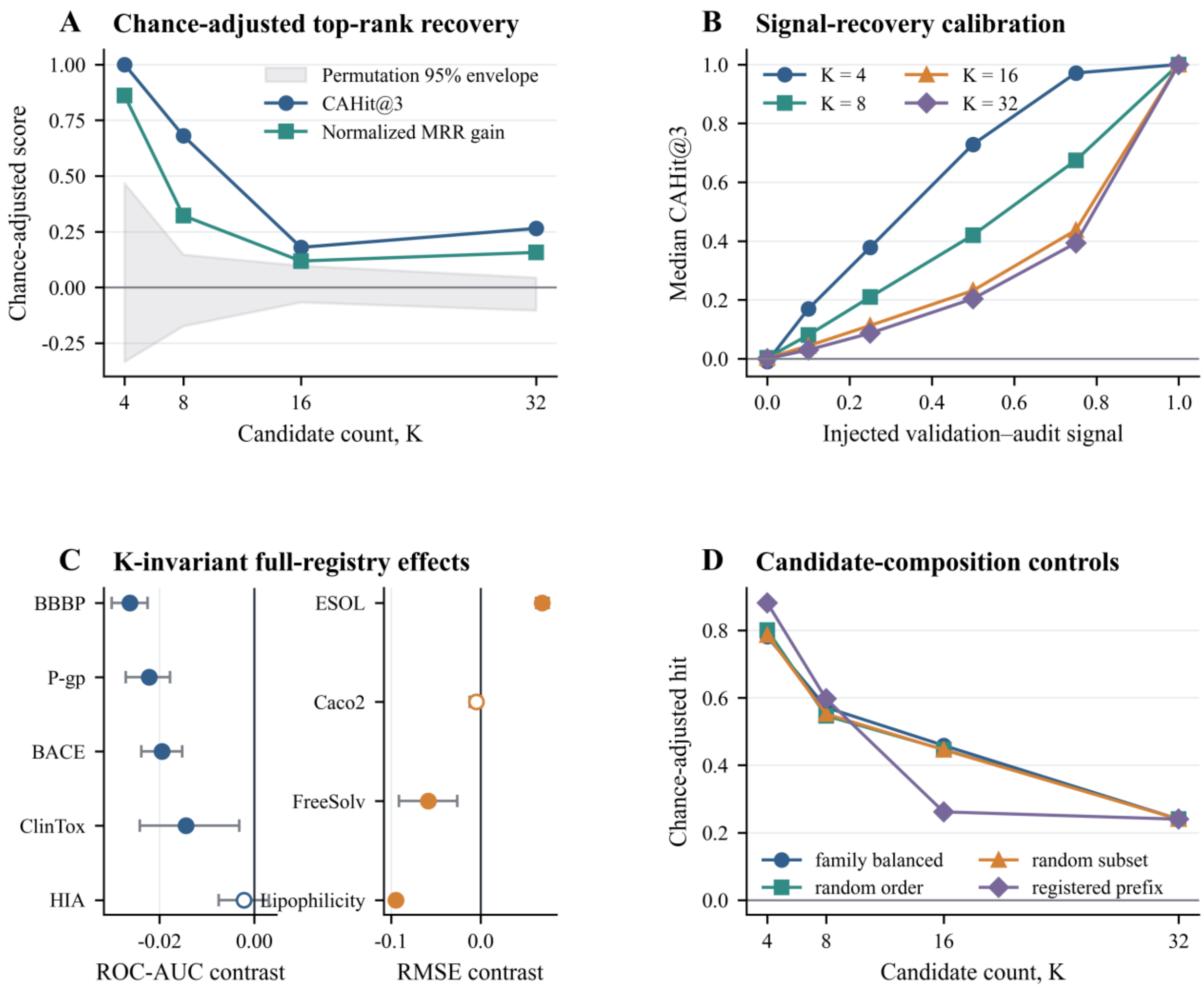

Figure 3. Chance-adjusted ranking calibration and K-invariant full-registry effects. (A) Five-seed secondary endpoint-median CAHit@3 and normalized MRR gain with endpoint IQRs and the 95% random-rank envelope. (B) Five-seed secondary positive-control recovery across six injected validation–audit signal levels and four candidate counts. (C) Ten-seed primary endpoint-specific K = 32 minus K = 4 full-registry completion-gap contrasts against the K-invariant K = 32 cross-fitted reference. Classification ROC-AUC and regression RMSE use separate axes; negative values indicate a smaller completion gap at K = 32, and filled markers denote split-seed sensitivity intervals that exclude zero. Panel C uses the same estimates and limits as Table 4. (D) Five-seed secondary registered-prefix, random-order, random-subset and family-balanced composition controls; all modes equal the complete registry at K = 32 by design.

3.4 K-invariant cross-fitted effects remained heterogeneous

Against the K-invariant K = 32 leave-one-seed-out reference, K = 32 minus K = 4 full-registry completion-gap contrasts were negative in 8 of 9 endpoints and positive in 1. A negative contrast means that the full-registry completion gap was smaller at K = 32; a positive contrast means that it was smaller at K = 4. Descriptive split-seed sensitivity intervals excluded zero for 7 endpoints. Figure 3C and Table 4 are generated from the same endpoint source table and retain the same estimates, interval limits and signs without pooling classification and regression scales.

Table 4. K-invariant full-registry completion-gap contrasts.

Endpoint	K = 32 minus K = 4 contrast (95% sensitivity interval)	Interpretation
Classification: ROC-AUC completion-gap contrast		
BACE	−0.0195 (−0.0238, −0.0152)	Smaller gap at K = 32; interval excludes 0
BBBP	−0.0263 (−0.0301, −0.0225)	Smaller gap at K = 32; interval excludes 0
ClinTox	−0.0144 (−0.0241, −0.0031)	Smaller gap at K = 32; interval excludes 0
HIA	−0.0022 (−0.0075, 0.0031)	Interval includes 0
P-gp	−0.0222 (−0.0271, −0.0178)	Smaller gap at K = 32; interval excludes 0
Regression: RMSE completion-gap contrast		
ESOL	0.0683 (0.0618, 0.0757)	Smaller gap at K = 4; interval excludes 0
FreeSolv	−0.0586 (−0.0912, −0.0263)	Smaller gap at K = 32; interval excludes 0
Lipophilicity	−0.0947 (−0.0992, −0.0899)	Smaller gap at K = 32; interval excludes 0
Caco2	−0.0049 (−0.0121, 0.0008)	Interval includes 0

Note: Negative values indicate a smaller completion gap relative to the full-registry reference at K = 32. Intervals are descriptive split-seed sensitivity intervals.

The K-invariant reference prevents the comparator identity from changing with K and separates its selection from the held-out seed. It remains conditional on the same public endpoint population and split generator, so it is a retrospective sensitivity reference rather than external validation or deployable truth.

3.5 Finite-audit maxima contained winner optimism

With pairwise candidate correlation 0.9 and effective audit size 50, mean winner optimism increased from 0.046 standard-deviation units at K = 4 to 0.092 at K = 32. At K = 32, it was 0.132 at effective audit size 25 and 0.047 at effective audit size 200. Lower candidate correlation produced larger finite maxima because less correlated candidate estimates provided more independent noise directions (Figure 4D).

This pattern is a mechanical property of finite maxima, not an empirical bias correction. Real candidates need not have equal population utilities, and endpoint-specific dependence can depart from equicorrelation. The simulation therefore does not estimate a correction for the empirical endpoints; it shows why the observed finite-audit best is not a population upper bound, even when the outer fold was untouched by inner selection.

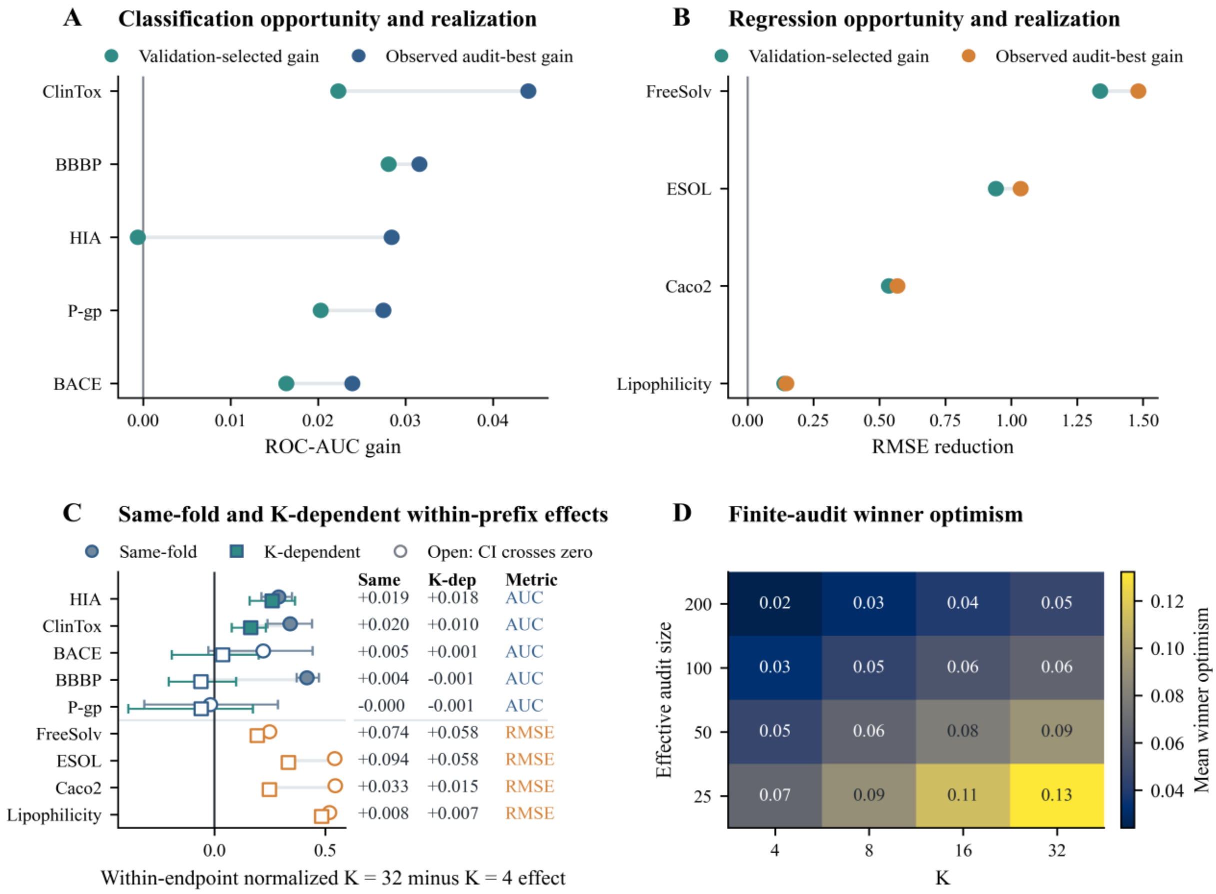

Figure 4. Selection-gap decomposition and winner optimism. Panels A and B compare observed audit-best and validation-selected gains for classification and regression. Panel C contrasts the same-fold finite-set opportunity gap with the K-dependent within-prefix cross-fitted gap as a secondary decomposition or sensitivity analysis; the primary K-invariant full-registry completion-gap estimate is shown in Figure 3C. For visualization across endpoints, panel C uses within-endpoint standardized effects; raw ROC-AUC and RMSE effects are listed separately. Panel D shows finite-audit winner optimism when candidate true utilities are equal.

3.6 Matched-size representation effects were endpoint dependent

The matched-size multiview audit retained substantial endpoint heterogeneity: composition distributions included positive, weak and negative effects, and increasing K did not eliminate those differences. The 220 K = 3 subsets overlap in candidate membership and are distributional contrasts within one registry, not independent experiments.

Classification ROC-AUC gains and regression RMSE reductions remain on separate axes. Complete subset identities, IQRs, ranges and ladder results are provided in Tables S9–S10 (Figure 5).

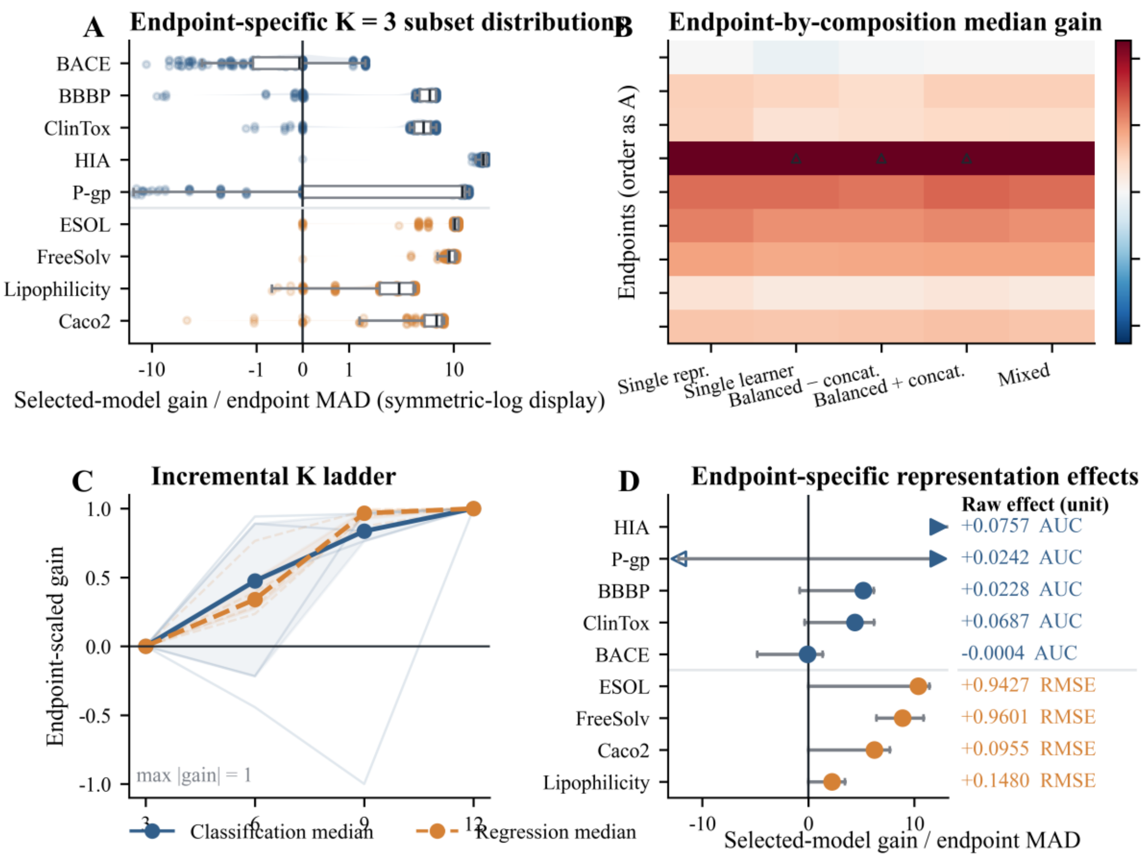

Figure 5. Matched-size representation-composition effects. Panel A shows endpoint-specific distributions across all 220 registered K = 3 subsets. Panel B summarizes mutually exclusive composition classes. Panel C shows the K = 3, 6, 9 and 12 ladder. Panel D presents all endpoints in one integrated forest plot using endpoint-MAD-normalized effects, with raw ROC-AUC gains and RMSE reductions listed separately.

3.7 Prediction errors were correlated but partially complementary

The limited four-model panel provided 90 endpoint-seed-fold prediction sets across six endpoints. Prediction correlations and error overlaps were substantial but incomplete; mean Ledoit–Wolf prediction entropy rank was lower than the nominal model count of four. These quantities characterize the completed fixed configurations and cannot rank fully optimized modern architectures.

Across chemical-support strata, pairwise high-error Jaccard overlap was non-zero and model disagreement increased at weaker support in several endpoints. Prediction-level complementarity therefore existed, but it was not equivalent to independent candidate opportunity. Unequal model-training budgets further restrict any architecture-level comparison.

The panel cannot support a performance hierarchy among adequately tuned modern methods. The GCN received only five fixed epochs, and both language-model representations used frozen linear probes rather than end-to-end fine tuning. RDKit-RF consequently had a different optimization budget and inductive bias. The completed outputs are informative for selector behaviour and error correlation under the evaluated configurations, but they should not be generalized to all graph networks, ChemBERTa variants or MoLFormer training strategies.

The support-stratified audit retained one traceable selected-candidate result per split-seed outer fold and avoided plotting duplicate selected and reference points. Candidate performance, error overlap and disagreement were summarized within prespecified support strata; no support stratum was used to choose a candidate (Figure 6A–B).

3.8 Reliability weakened at chemical-support boundaries

This analysis asked whether predictive reliability remained stable in sparse or discontinuous chemical regions. At 90% nominal coverage, mean minority-class coverage across classification endpoints was 0.788 for split conformal, 0.887 for label-conditional conformal and 0.891 for Mondrian label-and-similarity conformal prediction. Conditional methods substantially reduced minority undercoverage but remained slightly below the target on average. Prediction sets also became larger; thus, improved coverage was accompanied by reduced decisiveness.

CQR coverage was endpoint dependent: 0.836 for ESOL, 0.893 for FreeSolv and 0.917 for Lipophilicity at the 90% target. The cross-endpoint mean was 0.882. Raw interval widths were not pooled for interpretation because the targets use different units and dynamic ranges. Ensemble uncertainty had mean uncertainty–error Spearman correlation 0.321 and enriched the 10% of samples with the largest errors by 1.54-fold, indicating useful but incomplete error ranking.

Chemical-support stratification showed a boundary signal. Median classification ROC-AUC across audit units was 0.842 below Tanimoto 0.5, 0.914 from 0.5 to <0.7 and 0.925 at or above 0.7. Median regression RMSE was 1.337, 1.097 and 0.767 across the same strata. Novel-scaffold changes in error overlap, disagreement and high-error enrichment were expressed relative to seen-or-related scaffolds, which defined the unit reference (ratio = 1).

ClinTox remained the strongest negative result in the traceable four-model panel. Discrimination, minority recall, false-negative rate, label-conditional conformal coverage and mean prediction-set size are shown as distinct quantities rather than collapsed into one ranking. The observed minority-safety limitation precludes interpreting a favourable ROC-AUC as evidence of deployment readiness (Figure 6C–D).

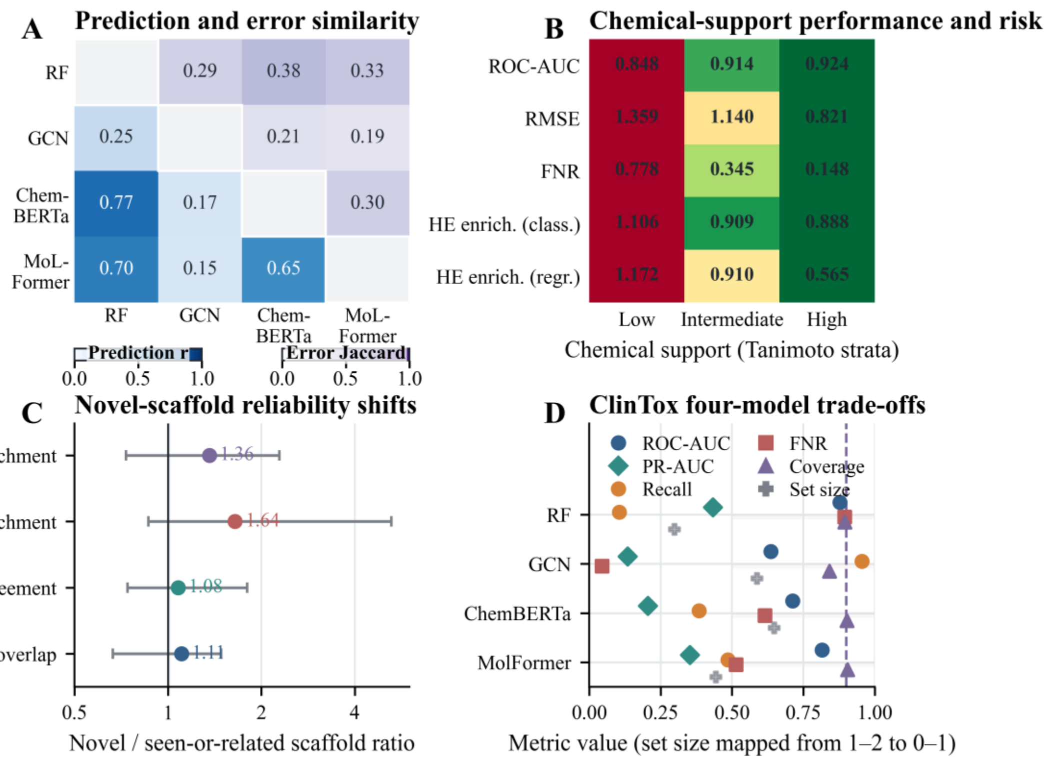

Figure 6. Prediction reliability across chemical-support boundaries. (A) Prediction correlations and high-error Jaccard overlaps are combined in a symmetric four-model matrix with separate in-panel colour strips. (B) Classification ROC-AUC, regression RMSE, classification false-negative rate and classification/regression high-error enrichment are consolidated into one Tanimoto-support risk matrix; cell text gives natural-scale medians and colour encodes only the within-row adverse direction. (C) Novel-scaffold relative changes in error overlap, model disagreement, high-error enrichment and false-negative enrichment are shown on a log-ratio axis. (D) Discrimination and minority-safety measures are separated for four fixed-configuration models; the purple dashed line is the 0.90 coverage target, and prediction-set size is mapped from its 1–2 scale only for display.

3.9 Audit directions transferred but effect magnitudes changed under structure separation

Across ClinTox, BACE and ESOL, the prespecified directions of CAHit@3 change and the cross-fitted gap were preserved from seeded scaffold partitions to Tanimoto-component splits, but effect magnitudes and several individual contrasts changed. Effective-rank ordering was partly retained while absolute values remained split dependent.

This three-endpoint transport analysis does not establish external validation, time-split robustness or a universal model-selection law. Complete intervals, splitter feasibility checks, missing cells and task-specific effects are retained in Tables S24–S26 and Figure S19.

3.10 Expanded composition, stability and compute controls

Across the six prespecified endpoints and three registries, CAHit@3 decreased from K = 4 to K = 32 in most endpoint–pool combinations, whereas cross-fitted gaps remained heterogeneous. At K = 32, multiview and modern augmentation produced positive validation-selected gains relative to the homogeneous registry across all six endpoints; raw classification and regression utilities were not pooled. Complete counts are reported in Table S33 and Figure S16.

Modern components were heterogeneous: frozen language-model representations, the separately locked one-epoch D-MPNN and the full modern registry did not contribute uniformly across endpoints or candidate-pool sizes. Component-level paired results are reported in Table S33 and Figure S17.

Equal-budget truncation and selection-stability analyses showed that realised benefit depended on the locked prefix order and the alignment between inner-validation ranking and outer-audit utility. Complete equal-K, equal-budget, selection-frequency and entropy results are reported in Table S36 and Figures S20–S21.

Anchor and normalization choices affected numerical scale more than the overall direction pattern. Most prespecified directions were preserved across the two evaluated split mechanisms, although their magnitudes changed. Missing cells and complete sensitivity results are retained in Tables S34–S35 and Figures S18–S19.

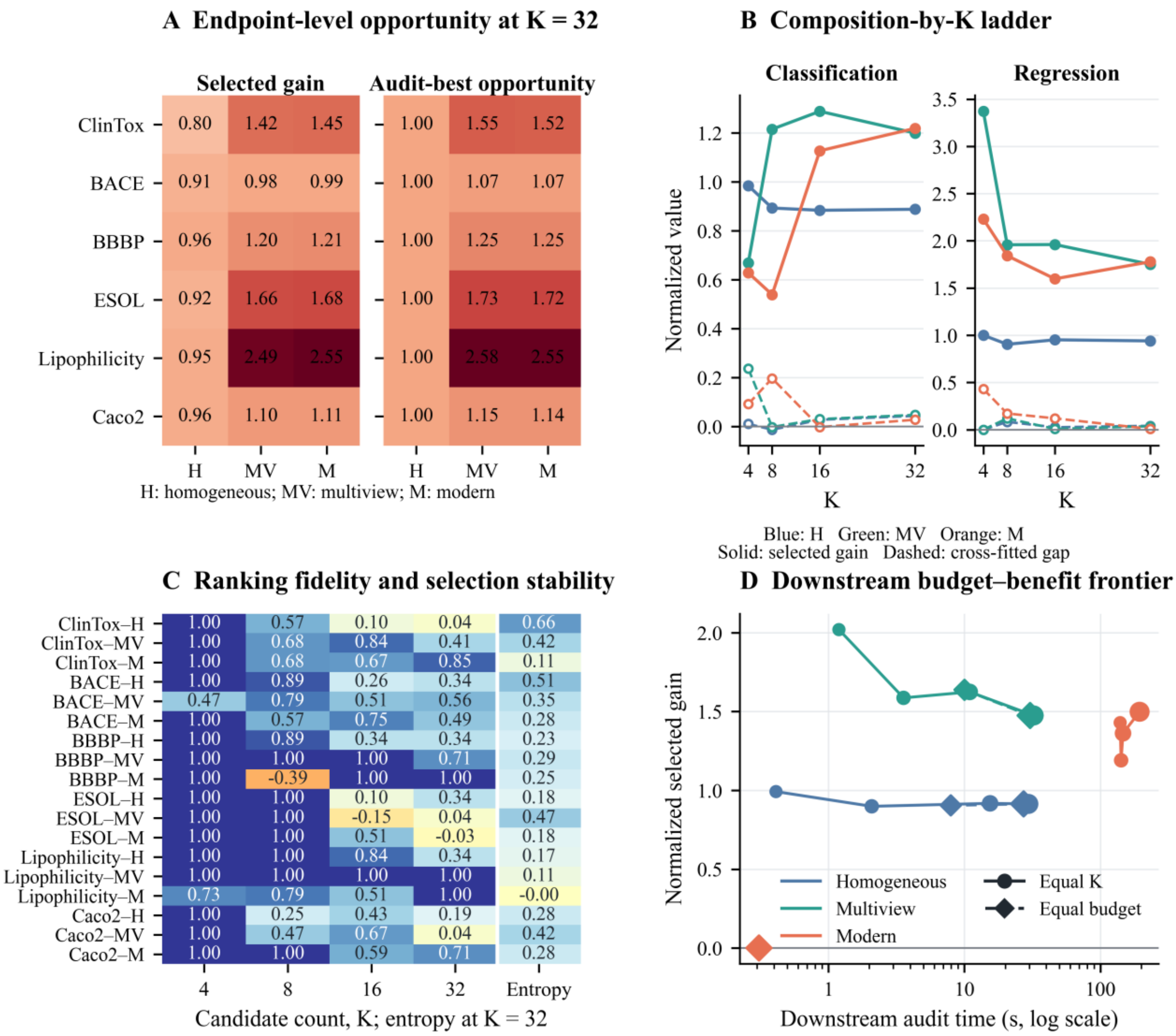

Figure 7. Expanded candidate-pool composition intervention. (A) At K = 32, normalized gain uses paired homogeneous-audit-best normalization: the validation-selected gain is divided by the corresponding homogeneous-pool finite-audit-best gain; selected and finite-audit-best opportunity values are shown for six prespecified endpoints. (B) Validation-selected gain and cross-fitted gap are shown at K = 4, 8, 16 and 32 in separate classification and regression task bands; the pool legend is shown once above panel B. (C) CAHit@3 is shown for each endpoint, pool and K combination, with negative values retained; the rightmost Entropy column in the main figure reports normalized candidate-selection entropy at K = 32. H denotes homogeneous Morgan, MV classical multiview and M modern-augmented. (D) Normalized selected gain and downstream fitting/prediction time are compared under equal-K circles and equal-downstream-budget diamonds; marker area denotes K and the empirical Pareto frontier is shown. This composition intervention is a five-seed secondary analysis after averaging three outer folds within seed; time excludes model acquisition, encoder pretraining and cached embedding extraction.

3.11 Practical equivalence and metric-dependent selection

Holding the K = 32 cross-fitted reference identity fixed across K removed comparator drift. At K = 32, classification τ-success was 62.7%, 80.0% and 86.0% across the 0.005/0.010/0.020 ROC-AUC grid; regression τ-success was 73.3%, 79.2% and 82.5% across the 0.025/0.050/0.100 RMSE grid. The pooled task-appropriate working-middle-threshold success rate was 79.6%; the mean cross-fitted set size was 10.17, the selected-in-set rate was 85.9% and mean validation–cross-fit set Jaccard overlap was 0.490. These are retrospectively locked reporting tolerances, not clinical or pharmacological thresholds.

At K = 32, PR-AUC selection changed the ROC-AUC-selected candidate in 52.7% of units and changed outer PR-AUC, ROC-AUC, minority recall, majority recall and false-negative rate by +0.0018, −0.0026, −0.0002, −0.0136 and +0.0002, respectively. The minority-recall-constrained selector changed candidates in 66.0% of units and changed the same outcomes by +0.0007, −0.0008, −0.0094, +0.0059 and +0.0094. An inner-eligible candidate existed in 100.0% of K = 32 units, whereas the frozen rule achieved outer minority recall ≥0.80 in 62.0%; the negative mean recall change was retained rather than interpreted as automatic constraint transfer (Tables S38–S40).

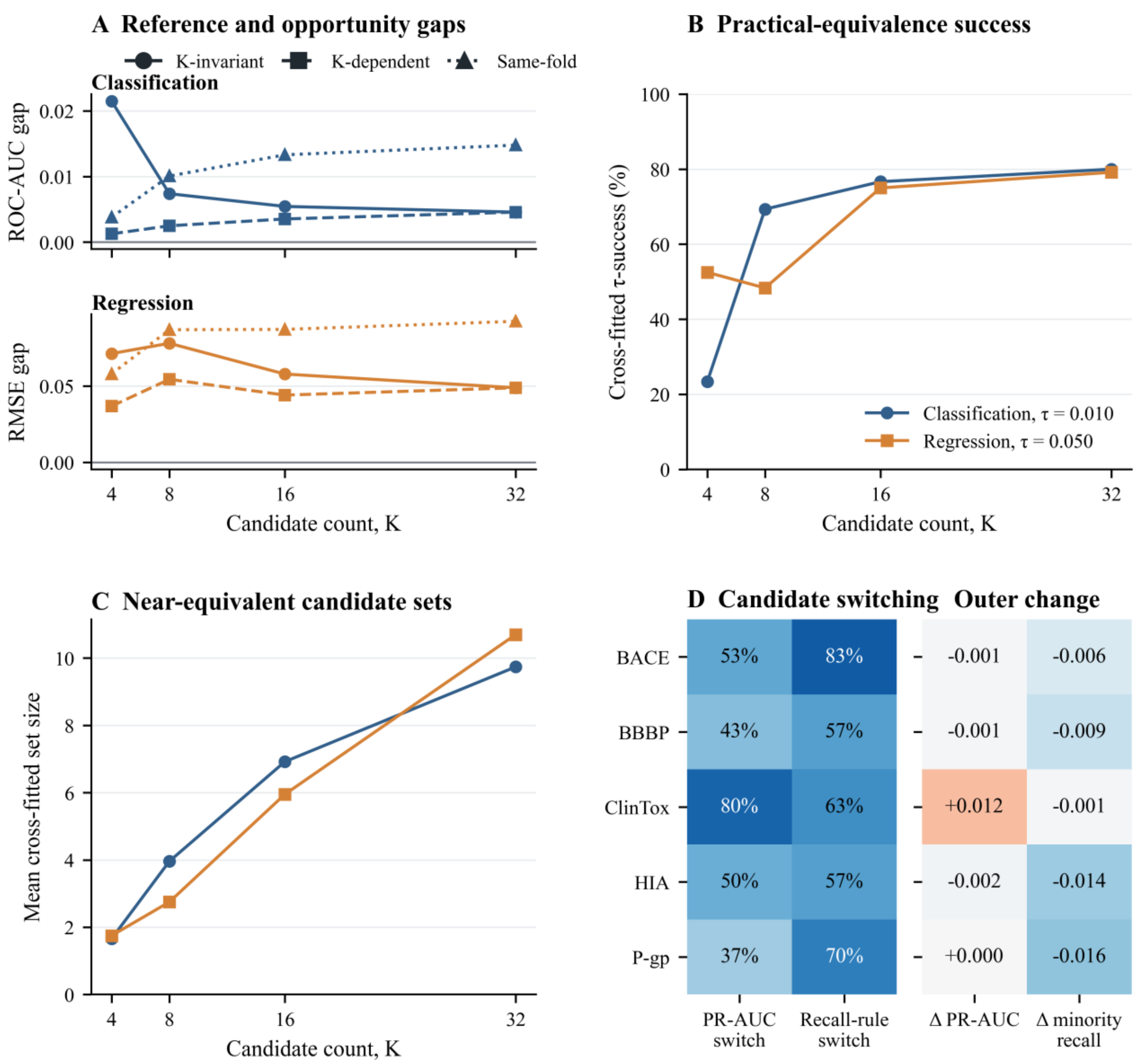

Figure 8. Practical equivalence and metric-dependent selection in the ten-seed primary audit. (A) Classification ROC-AUC and regression RMSE gaps are shown on separate stacked axes sharing candidate count K; circles/solid lines, squares/dashed lines and triangles/dotted lines denote K-invariant, K-dependent and same-fold estimands, respectively, so estimands remain distinguishable in greyscale. (B) Cross-fitted τ-success is shown across K at the retrospectively locked middle practical-equivalence tolerances τ. (C) Mean cross-fitted near-equivalent set size is shown on one axis; the classification/regression legend in panel B also applies to panel C. (D) Candidate switching and outer changes are shown for the five classification endpoints, with PR-AUC switch, recall-rule switch, ΔPR-AUC and Δminority recall labelled explicitly; negative minority-recall changes are retained. Panels A–D are arranged in a 2 × 2 layout.

3.12 Constructed dependence controls and empirical recovery

Exact duplication increased nominal K without adding unique utility information, whereas near-duplicate, weak-first and complementary-first constructions changed effective rank and opportunity differently (Figure S25; Table S41). Under the empirical correlated heteroskedastic simulation at K = 32 and the estimated noise scale, mean top-1 recovery across endpoints was 0.327; observed mean same-fold gaps fell within the simulation's 95% envelope for 3 of 9 endpoints (Figure S23; Table S42). These constructed and simulated analyses calibrate mechanisms and are neither empirical bias corrections nor independent validation.

3.13 Secondary contextual and negative analyses

The five-seed secondary supplement also retained results that bound, rather than confirm, the ten-seed primary estimand. Twenty-two TDC endpoint summaries used heterogeneous historical budgets and were treated as contextual reliability evidence; AutoGluon results at 30, 300 and 1,800 seconds had unequal feature-preparation and hardware exposure; selection-risk correlations and nine leave-one-endpoint-out policies could not identify a transferable meta-selector. Conformal and CQR analyses improved conditional coverage in some cells while enlarging prediction sets, and MoleculeACE activity-cliff, beyond-rule-of-five, deduplication and failure-case panels exposed chemical-support and minority-safety limitations. These supplementary analyses are described in Sections S8–S13 and Tables S13–S20; none was pooled with the 34,560-fit ten-seed primary matrix.

4 Discussion

4.1 Nominal K did not define candidate diversity

Candidate count, family count, representation count, compute exposure and effective diversity describe different aspects of a model search. Nominal K records how many candidates were eligible, whereas family and representation counts describe registry composition and compute exposure records the fitting effort expended. None of these counts establishes how independently candidates behaved across audit units.

The registered prefixes increased candidate and compute exposure mechanically, but their candidates shared representations and closely related learners. Effective diversity instead summarized the variation in observed utility patterns and remained conditional on the evaluated endpoints and splits. Molecular benchmark reports should therefore distinguish search size, registry composition, computational exposure and matrix-dependent effective diversity rather than treating any one quantity as a substitute for the others.

4.2 Matrix construction changed effective-rank interpretation

Raw utility, row-centred utility, fixed-reference-relative utility and within-unit rank matrices answer different questions. Raw utilities retain common audit-unit difficulty; row-centred utilities remove common level shifts while imposing a row-sum constraint; fixed-reference contrasts depend on the prespecified reference; and within-unit ranks preserve ordering while discarding utility spacing. Their effective ranks should not be interpreted as competing estimates of one intrinsic candidate count.

In the primary ten-seed analysis, each endpoint supplied 30 outer audit units. Because the largest candidate count $K = 32$ still slightly exceeded the number of audit units, the unshrunk empirical candidate-correlation matrix could remain rank deficient. Ledoit–Wolf shrinkage stabilized covariance directions but did not create new independent information. Spectral-entropy and participation-ratio ranks weight the eigenvalue spectrum differently, so both were reported with omission and prespecified-reference sensitivities.

4.3 Chance-adjusted ranking discordance accompanied selection gaps

Raw winner recovery becomes mechanically harder as K grows, so the interpretation relies on opportunity-adjusted and graded ranking measures. CAHit@3 asks whether the observed finite-audit best entered a short validation list, normalized MRR credits its position relative to the random-order expectation, NDCG emphasizes the ordered utility profile, and Spearman, Kendall and rank percentile capture broader reordering.

The permutation negative control placed both chance-adjusted metrics near their random-order zero points, and the observed endpoint median CAHit@3 at every K exceeded the corresponding 95% permutation envelope. In the graded positive control, CAHit@3 and normalized MRR increased monotonically, while fixed-range selection loss decreased monotonically and reached zero under perfect signal as the injected validation–audit signal increased. These prespecified simulations confirmed the expected zero points and signal-recovery behaviour but cannot establish a causal relationship between any single ranking metric and selection loss in real endpoints.

4.4 Cross-fitting separated the primary comparison from the same-fold opportunity bound

A same-fold maximum reuses the evaluated outer scores to define the reference and can therefore overstate the opportunity available to the selector. The leave-one-seed-out cross-fitted reference separated reference selection from held-out-seed evaluation and attenuated same-fold effects in most endpoints while retaining positive and negative cases. We therefore interpret the cross-fitted gap as the principal retrospective comparison and the same-fold gap only as a finite-set opportunity bound.

Cross-fitting did not remove finite-maximum optimism, model-registry conditioning or public endpoint reuse. The known-truth simulation illustrated finite-maximum optimism when noisy candidate estimates were maximized, but it was not used as a bias correction. Because the cross-fitted reference still uses the same public endpoint population and split generator, it is sensitivity evidence rather than external validation. Reporting cross-fitted and same-fold quantities together makes dependence on the evaluated-unit maximum visible without treating either as an absolute truth.

4.5 Matched-size analyses refined the multiview interpretation

Exhaustive enumeration of 220 overlapping matched-size subsets held candidate count fixed while changing representation and learner composition. These subsets characterize sensitivity within one registry and are not independent experiments. Mutually exclusive composition classes prevent balance, concatenation, representation and learner labels from being interpreted as independent factors.

Endpoint-specific distributions retained heterogeneous and negative results, including the negative median for BACE. ROC-AUC gains and RMSE reductions were not pooled on a common scale. Learner interaction and representation breadth remained intertwined in the larger K ladder, and matched size did not equalize every tuning or training budget. The analysis therefore supports endpoint-dependent composition effects rather than a general claim that multiview expansion is beneficial.

4.6 Reliability remained conditional on chemical support

Tanimoto support and novel-scaffold analyses showed that selection results remained conditional on chemical neighbourhood. Evidence from activity cliffs further demonstrates that close structural similarity does not guarantee similar properties, so applicability cannot be summarized by one average scaffold score. Error overlap, disagreement and support-stratified performance provide complementary boundary evidence.

ClinTox illustrates why discrimination, conformal coverage and screening safety must remain separate. ROC-AUC can coexist with weak rare-class behaviour, and nominal coverage does not guarantee acceptable minority recall or false-negative risk. Conditional methods may improve coverage by enlarging prediction sets, but they do not establish deployment readiness. These reliability analyses delimit the model-selection audit rather than independently confirm it.

4.7 Split-regime transfer reduced, but did not remove, design dependence

Preserving the CAHit@3 change direction and the cross-fitted gap direction under a split that prohibited cross-fold Tanimoto of 0.70 or greater argues against attributing the primary finding solely to ordinary scaffold allocation. The rank correlation of matrix-dependent diversity estimates also shows that broad registry structure was partly preserved when the split mechanism changed.

The transfer was conditional rather than invariant. BACE showed almost no CAHit@3 change under the similarity-cluster split, normalized MRR was not uniformly lower, and interval exclusion shifted across endpoints. Split mechanism therefore changed both difficulty and the estimated magnitude of selection pressure. The appropriate claim is direction transport in three representative endpoints, not a universal law across targets, temporal splits, model registries or deployment populations.

4.8 Registry composition, selection stability and bounded compute

Registry composition changes the opportunity available to a selector, whereas finite validation influences how much of that opportunity is realised. Candidate expansion can therefore raise observed audit-best utility and selection pressure simultaneously. Matrix-dependent effective diversity complements nominal K by describing relative independence of candidate utilities across audit units, but it is descriptive and does not itself cause or determine selection loss.

Modern candidates were not a homogeneous class: frozen language-model representations, the separately locked one-epoch D-MPNN and the full modern registry affected endpoints differently. This heterogeneity explains why modern augmentation can improve the opportunity frontier without guaranteeing that the inner selector captures every gain.

Equal-budget conclusions depend on the prespecified prefix order because an expensive early candidate can consume the downstream budget before later complementary candidates become eligible. Anchors and normalization methods affected numerical scale more than direction. These findings concern bounded downstream fit/predict exposure and do not imply end-to-end architecture efficiency within or beyond the evaluated endpoints, registries and split mechanisms.

4.9 Practical equivalence and metric-matched selection

Near-equivalent sets prevent negligible candidate differences from being interpreted as severe selection failures. However, the working τ values were retrospectively locked practical-equivalence reporting tolerances rather than endpoint-specific clinical or pharmacological thresholds. PR-AUC and minority-recall-constrained selection exposed genuine candidate-switching and safety-performance trade-offs, but the 0.80 target is a study rule and must be redefined for deployment costs.

4.10 Candidate dependence and finite validation

Exact and near-duplicate controls show that nominal K can grow while effective utility information changes little. Weak and complementary orderings show that composition and exposure order can alter both available opportunity and selector burden. The empirical residual simulation recovered observed gaps for some but not all endpoints, supporting finite-validation noise as one mechanism without reducing endpoint heterogeneity to a single mechanical explanation.

4.11 Limitations

This study is a retrospective audit of public datasets, not a prospective study, independent external validation or deployment evaluation. The primary analysis contains ten split seeds and three outer folds per seed, but its uncertainty information mainly comes from ten seed blocks; resampling cannot increase the independent information supplied by those blocks.

The K-invariant full-registry reference separates reference selection from the held-out seed, but it reuses the same public endpoints, candidate registry and split generator. It therefore reduces one source of circularity without replacing validation in an independent cohort or data source.

The practical-equivalence tolerances τ are retrospectively locked reporting tolerances rather than clinical, pharmacological or industrial minimum important differences. PR-AUC and training-defined minority-recall rules changed the inner objective, but the corresponding outer-fold constraints did not transfer reliably.

Coverage of modern candidates was limited. Equal-budget analyses record downstream fitting and prediction time but exclude model acquisition, encoder pretraining, cached embedding extraction and complete end-to-end cost. Tanimoto-component transport covered only three endpoints and one similarity rule.

Endpoints, seeds, folds, candidates and overlapping subsets are dependent. Constructed duplicate, near-duplicate, weak and complementary controls and empirical recovery simulations support a mechanism-based interpretation, but they do not establish a single causal effect of effective diversity on selection gaps.

5 Conclusions

Within the evaluated endpoints, candidate registries and split mechanisms, candidate-pool expansion altered model opportunity, validation–audit ranking agreement, practical equivalence, metric-dependent selection and bounded downstream cost. These changes depended on registry composition, candidate dependence, chemical support and endpoint. Nominal K and utility-pattern diversity described different search exposures; neither alone determined the K-invariant cross-fitted gap.

Molecular property-prediction benchmarks should report locked candidate eligibility, K-invariant and K-dependent cross-fitted references, the descriptive same-fold opportunity bound, near-equivalent sets, task-matched selection metrics, constructed dependence controls, selection stability, computational exposure and chemical-support boundaries. The present audit does not support universal claims that more candidates are necessarily worse or that any architecture class is generally superior.

Supplementary Information

Additional file 1 (PDF; .pdf). Title: Supplementary Methods and Results. Description: Supplementary evidence hierarchy, original analyses and completion methods/results for K-invariant references, practical equivalence, metric-matched selection, constructed controls, order sensitivity, recovery simulation and execution audit; Tables S1–S46 and Figures S1–S25 are cross-referenced.

Additional file 2 (XLSX; .xlsx). Title: Machine-readable Supplementary Tables S1–S46. Description: Original machine-readable tables plus fixed-reference units, ε sets, classification metric selection, constructed candidate controls, recovery simulation, order sensitivity, fit audit, pretraining-overlap status and the legacy-to-current rerun/split-transition audit.

Additional file 3 (PDF; .pdf). Title: Supplementary Figures S1–S25. Description: Original supplementary figures plus fixed-reference/equivalence, recovery, metric-selection and constructed-dependence figures.

Additional file 4 (ZIP; .zip). Title: Code and reproducibility package. Description: Source code, locked configurations, compact source results, manifests, equation-to-code mapping, integrity hashes and rerun entry points. Its portable base corresponds to release paper-release-2026-07-r9 and commit 9635a902fa3cc7bb7b71a234c1b2bbbe415193f0; the paper43 completion overlay is post-release and must be mirrored to a synchronized public release before submission.

List of abbreviations

BACE, beta-secretase 1; BBBP, blood–brain barrier penetration; CAHit@3, chance-adjusted Top-3 hit; CQR, conformalized quantile regression; D-MPNN, directed message-passing neural network; ECE, expected calibration error; HIA, human intestinal absorption; MACCS, Molecular ACCess System; MRR, mean reciprocal rank; NDCG, normalized discounted cumulative gain; P-gp, P-glycoprotein; PR-AUC, precision–recall area under the curve; RMSE, root mean squared error; ROC-AUC, receiver operating characteristic area under the curve.

Declarations

Ethics approval and consent to participate

Not applicable. This study used publicly available molecular datasets and involved no human participants, human data or animal experiments.

Consent for publication

Not applicable.

Availability of data and materials

The datasets supporting this article are public; source details are listed in Additional file 2, Table S1. Processed audit tables and reproducibility materials are included in Additional files 1–4.

Verified public release: https://github.com/zfr0857/FZYC-Mol/releases/tag/paper-release-2026-07-r9.

Immutable commit: 9635a902fa3cc7bb7b71a234c1b2bbbe415193f0.

Additional file 4 contains that portable base plus the post-release paper43 ten-seed completion overlay. Because the overlay is not present in a verified public r10 tag, repository synchronization remains a submission blocker.

No separate archival DOI was available at the time of this review.

Software record. Project name: FZYC-Mol candidate-pool audit. Project home page: https://github.com/zfr0857/FZYC-Mol. Archived version: paper-release-2026-07-r9, commit 9635a902fa3cc7bb7b71a234c1b2bbbe415193f0. Operating system: platform independent where the locked Python environment is supported. Programming language: Python 3.13.7. Licence: MIT. Restrictions on reuse: none beyond the licence and the source-dataset terms.

Use of generative artificial intelligence

During manuscript preparation and software/document quality control, the authors used OpenAI Codex to assist with code review, language editing, document formatting and consistency checks. The authors reviewed and verified all generated content and take full responsibility for the final manuscript. The tool was not listed as an author and did not determine the reported experimental results. The reporting and release workflow follows reproducibility guidance for cheminformatics research [32]. The submission-version code, configurations and integrity manifests are provided in Additional file 4.

Competing interests

[Not provided — CONFLICT HOLD; no declaration inferred.]

Funding

[Not provided — CONFLICT HOLD; no declaration inferred.]

Authors' contributions

[Not provided — CONFLICT HOLD; no declaration inferred.]

Acknowledgements

[Not provided — CONFLICT HOLD; no declaration inferred.]

References

1. Wu Z, Ramsundar B, Feinberg EN, et al. MoleculeNet: a benchmark for molecular machine learning. Chem Sci. 2018;9:513–530. doi:10.1039/C7SC02664A.
2. Huang K, Fu T, Gao W, et al. Therapeutics Data Commons: machine learning datasets and tasks for drug discovery and development. NeurIPS Datasets and Benchmarks Track. 2021. arXiv:2102.09548.
3. Cawley GC, Talbot NLC. On over-fitting in model selection and subsequent selection bias in performance evaluation. J Mach Learn Res. 2010;11:2079–2107.
4. Varma S, Simon R. Bias in error estimation when using cross-validation for model selection. BMC Bioinformatics. 2006;7:91. doi:10.1186/1471-2105-7-91.
5. Zhao D, Zhu Y, Wu Z, et al. Revisiting ADMET prediction reliability under real-world challenges in the foundation model era. J Cheminform. 2026;18:95. doi:10.1186/s13321-026-01217-2.
6. Zhang L, Zeng Y, Qi Y, et al. DCPM-ADMET: fusion of dual-component pre-trained model and molecular fingerprints to enhance drug ADMET properties prediction. J Cheminform. 2026. doi:10.1186/s13321-026-01244-z.

7. Jang Y, Lee J, Jeong K, Kim J. Multimodal graph fusion with statistically guided parsimonious descriptor selection for molecular property prediction. J Cheminform. 2026;18:18. doi:10.1186/s13321-025-01140-y.

8. Zhang Y, Liu W, Zhao H, et al. MolGramTreeNet: a multimodal molecular property prediction model via grammar tree-constrained molecular representation. iScience. 2026;29:114928. doi:10.1016/j.isci.2026.114928.

9. Wen X, Liu H, Long W, Wei S, Zhu R. Consistent semantic representation learning for out-of-distribution molecular property prediction. Brief Bioinform. 2025;26:bbaf147. doi:10.1093/bib/bbaf147.

10. Yin T, Gao P, Panapitiya G, Saldanha EG. Out-of-distribution evaluation of active learning pipelines for molecular property prediction. RSC Adv. 2026;16:5281–5295. doi:10.1039/D5RA08055J.

11. Uchibori Y, Kaneko H. Generation of molecules near the applicability domain boundaries of property prediction models. J Chem Inf Model. 2026;66:6866–6879. doi:10.1021/acs.jcim.5c03220.

12. Kim JY, Vlachos DG. Distance-aware molecular property prediction in nonlinear structure–property space. J Chem Inf Model. 2025;65:6744–6756. doi:10.1021/acs.jcim.5c01037.

13. Tang H, Yue T, Li Y. Assessing uncertainty in machine learning for polymer property prediction: a benchmark study. J Chem Inf Model. 2025;65:6585–6598. doi:10.1021/acs.jcim.5c00550.

14. Fralish Z, Reker D. Pairwise learning for molecular property prediction and optimization. Front Drug Discov. 2026;6:1859068. doi:10.3389/fddsv.2026.1859068.

15. Landrum G. RDKit: open-source cheminformatics software. Version 2026.3.1. https://www.rdkit.org/. Accessed 12 July 2026.

16. Rogers D, Hahn M. Extended-connectivity fingerprints. J Chem Inf Model. 2010;50:742–754. doi:10.1021/ci100050t.

17. Breiman L. Random forests. Mach Learn. 2001;45:5–32. doi:10.1023/A:1010933404324.

18. Ke G, Meng Q, Finley T, et al. LightGBM: a highly efficient gradient boosting decision tree. Adv Neural Inf Process Syst. 2017;30.

19. Chen T, Guestrin C. XGBoost: a scalable tree boosting system. Proc ACM SIGKDD. 2016:785–794. doi:10.1145/2939672.2939785.

20. Prokhorenkova L, Gusev G, Vorobev A, Dorogush AV, Gulin A. CatBoost: unbiased boosting with categorical features. Adv Neural Inf Process Syst. 2018;31:6638–6648.

21. Yang K, Swanson K, Jin W, et al. Analyzing learned molecular representations for property prediction. J Chem Inf Model. 2019;59:3370–3388. doi:10.1021/acs.jcim.9b00237.

22. Chithrananda S, Grand G, Ramsundar B. ChemBERTa: large-scale self-supervised pretraining for molecular property prediction. arXiv:2010.09885. 2020.

23. Ross J, Belgodere B, Chenthamarakshan V, et al. Large-scale chemical language representations capture molecular structure and properties. Nat Mach Intell. 2022;4:1256–1264. doi:10.1038/s42256-022-00580-7.

24. Erickson N, Mueller J, Shirkov A, et al. AutoGluon-Tabular: robust and accurate AutoML for structured data. arXiv:2003.06505. 2020.

25. Tropsha A. Best practices for QSAR model development, validation, and exploitation. Mol Inform. 2010;29:476–488. doi:10.1002/minf.201000061.

26. Vovk V, Gammerman A, Shafer G. Algorithmic Learning in a Random World. New York: Springer; 2005. doi:10.1007/b106715.

27. Shafer G, Vovk V. A tutorial on conformal prediction. J Mach Learn Res. 2008;9:371–421.

28. Guo C, Pleiss G, Sun Y, Weinberger KQ. On calibration of modern neural networks. Proc ICML. 2017;70:1321–1330.

29. van Tilborg D, Alenicheva A, Grisoni F. Exposing the limitations of molecular machine learning with activity cliffs. J Chem Inf Model. 2022;62:5938–5951. doi:10.1021/acs.jcim.2c01073.

30. Sheridan RP. Time-split cross-validation as a method for estimating the goodness of prospective prediction. J Chem Inf Model. 2013;53:783–790. doi:10.1021/ci400084k.

31. Demšar J. Statistical comparisons of classifiers over multiple data sets. J Mach Learn Res. 2006;7:1–30.

32. Hoyt CT, Zdrazil B, Guha R, et al. Improving reproducibility and reusability in the Journal of Cheminformatics. J Cheminform. 2023;15:62. doi:10.1186/s13321-023-00730-y.

33. Parrondo-Pizarro R, Lanini J, Rodríguez-Pérez R. Uncertainty quantification in molecular machine learning for property predictions under data shifts. J Chem Inf Model. 2026;66:923–935. doi:10.1021/acs.jcim.5c02381.

34. Deng J, Yang Z, Wang H, Ojima I, Samaras D, Wang F. A systematic study of key elements underlying molecular property prediction. Nat Commun. 2023;14:6395. doi:10.1038/s41467-023-41948-6.

35. Li Z, Chen X, Wen H, et al. A systematic survey and benchmark of deep learning for molecular property prediction in the foundation model era. arXiv:2604.16586. 2026.

36. Romano Y, Patterson E, Candès EJ. Conformalized quantile regression. Adv Neural Inf Process Syst. 2019;32.

37. FZYC-Mol Authors. FZYC-Mol: candidate-pool expansion audit code and reproducibility package. Version paper-release-2026-07-r9 (commit

9635a902fa3cc7bb7b71a234c1b2bbbe415193f0). GitHub; 2026. Available from: https://github.com/zfr0857/FZYC-Mol. Accessed 22 July 2026.